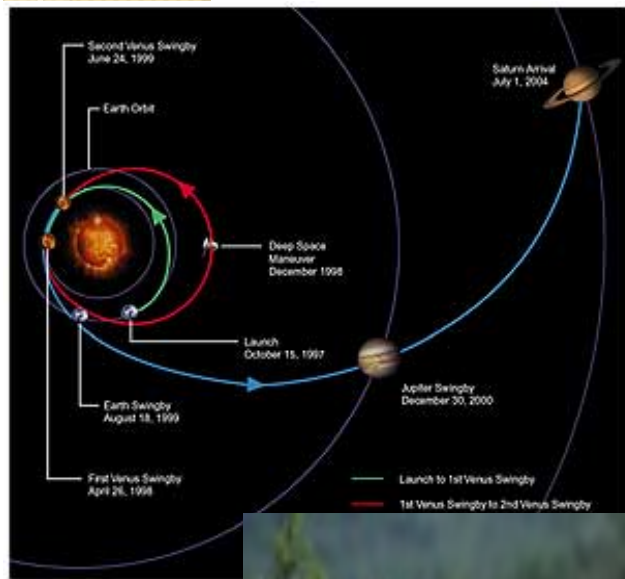




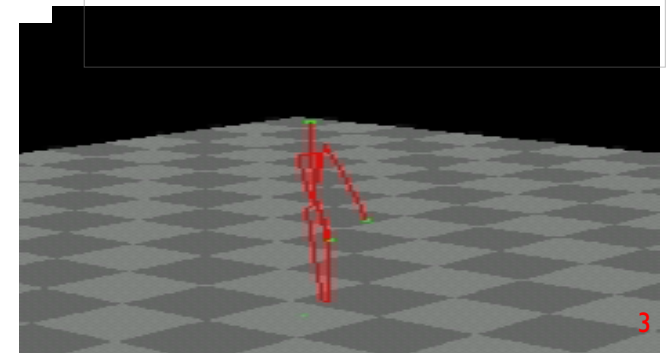
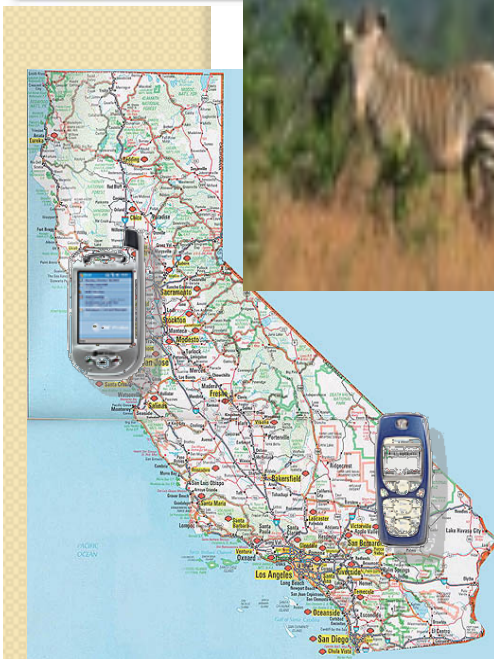
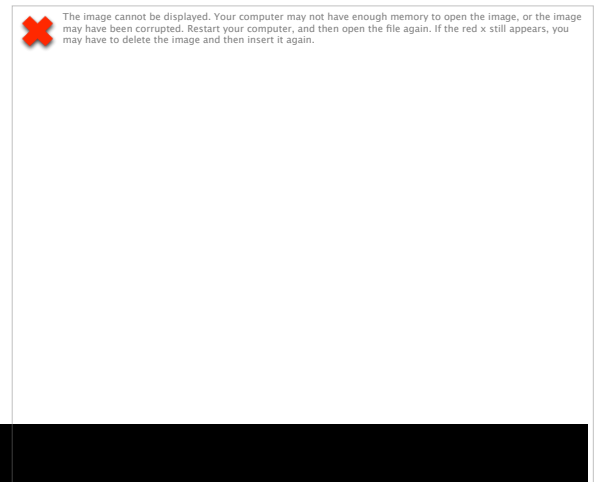
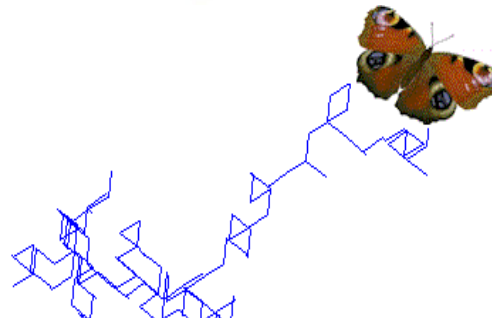
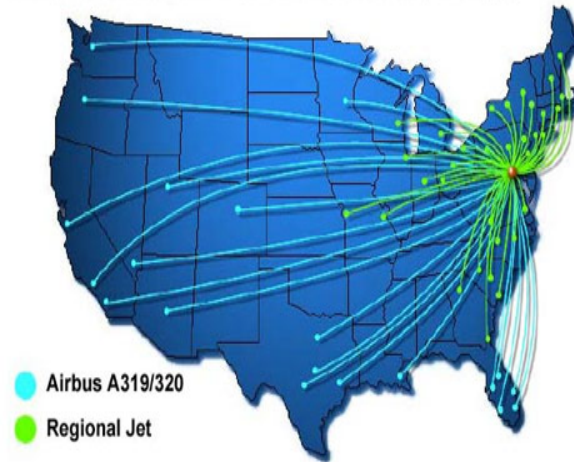
Similarity of Spatio-Temporal Datasets

Dimitrios Gunopulos
University of Athens
dg@di.uoa.gr

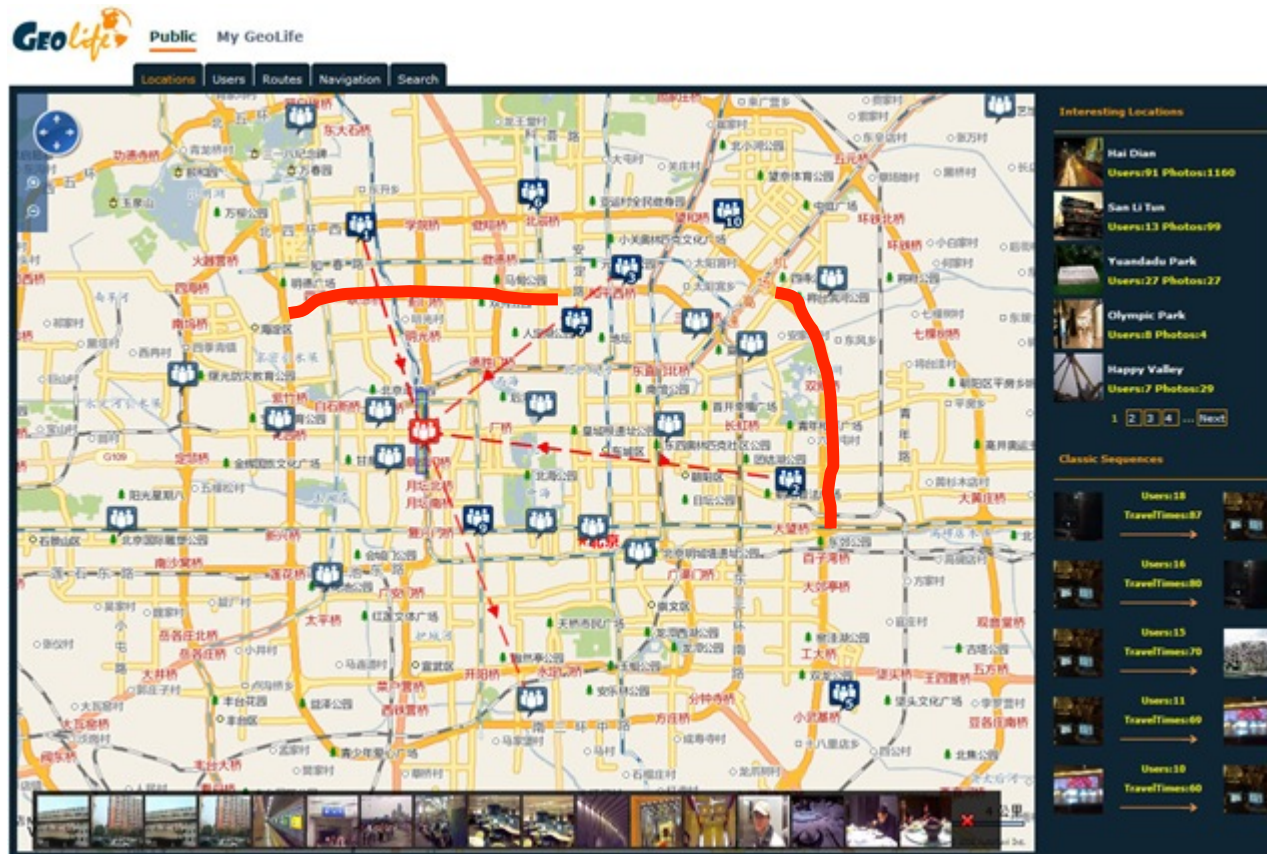
Trajectories are everywhere



MAPA LINEK INDEPENDENCE AIR V ROCE 2006



- Plethora of Application Domains:
 - Traffic Management

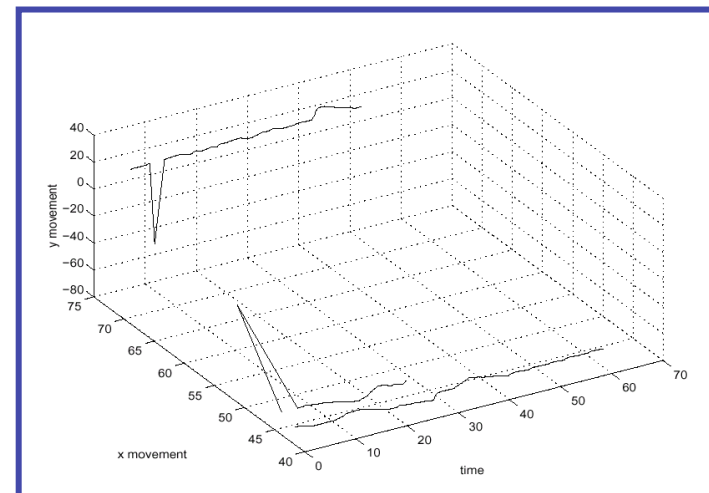
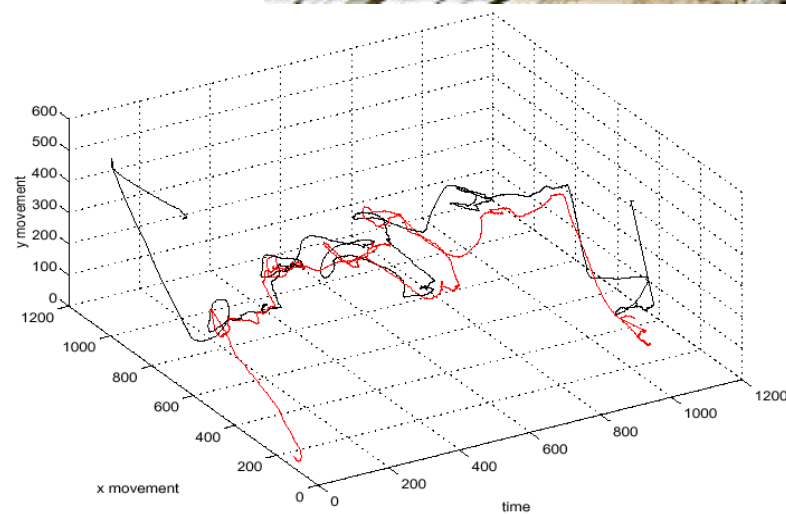


Retrieve all the ambulances which had **similar delays** patterns during the rush hours in the last 6 months

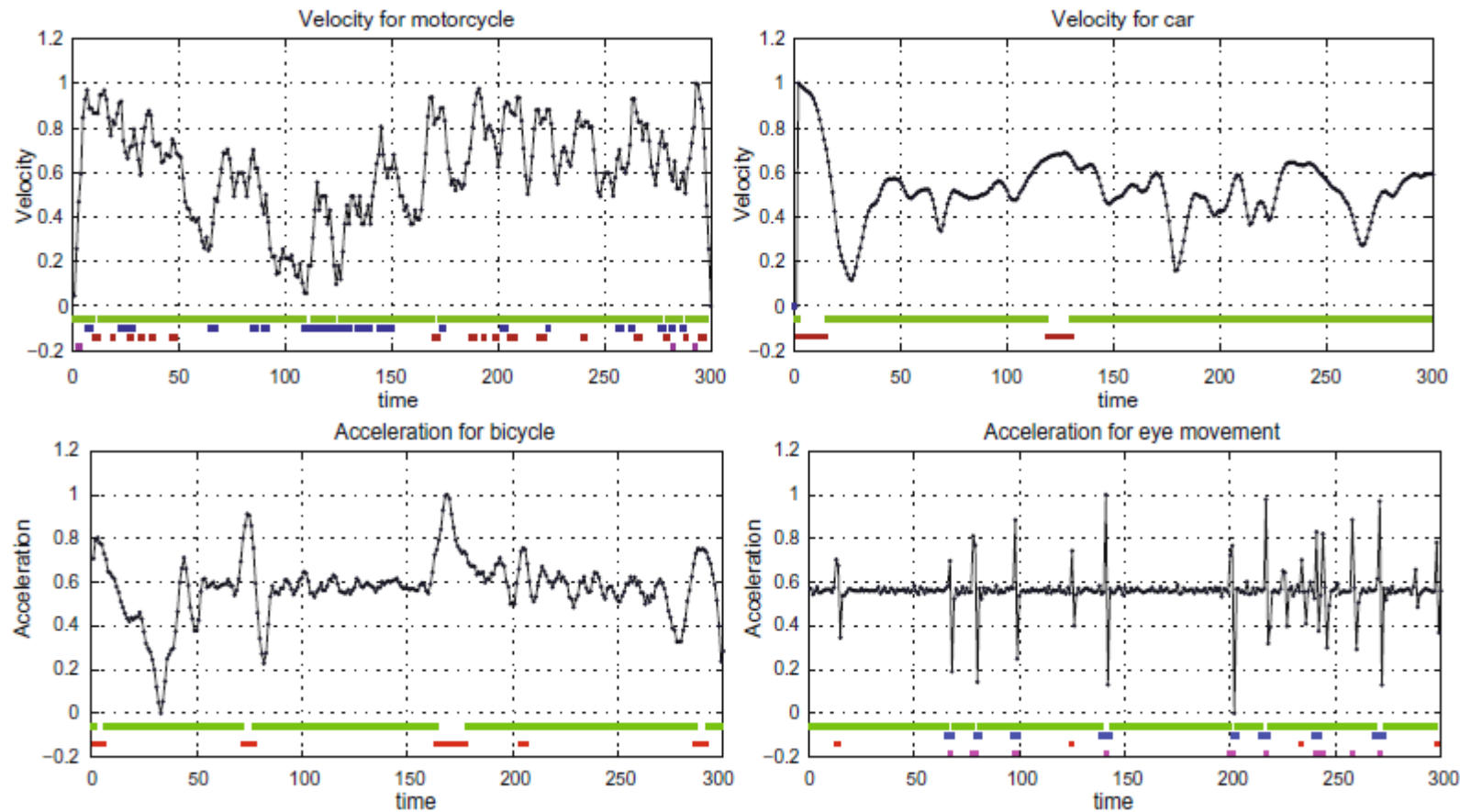


Multi-Dimensional Time-Series

Mobile usage,
Tracking,
Video, Motion
Capture,
Games



- Plethora of Application Domains:
 - Similarity of Motions in Different Contexts – what should we use as a “measure”(?)



Similar Sinuosity...

- Moving Object

Changes its spatial whereabouts over time
(during its existence)

Any real-world element
That may be perceived as
“unique” (a car, a person)

Trajectory: continuous mapping from Time to (some) Geographical Space
 $I(\subseteq) \mathbb{R} \rightarrow \mathbb{R}^2$

or, even think of it as parameterization over time

$$\alpha(t) = (\alpha_x(t), \alpha_y(t))$$

So, now

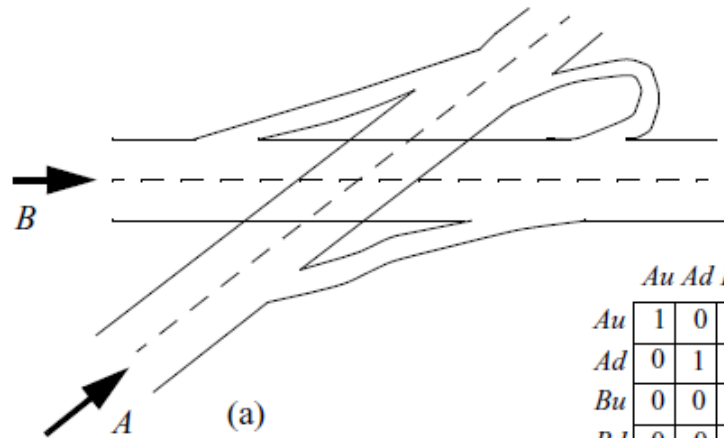
Trajectory = $\{(\alpha_x(t), \alpha_y(t), t) \mid t \in I\}$

- Description
- Representation
- Manipulations

[MVO+08, GBE+00]

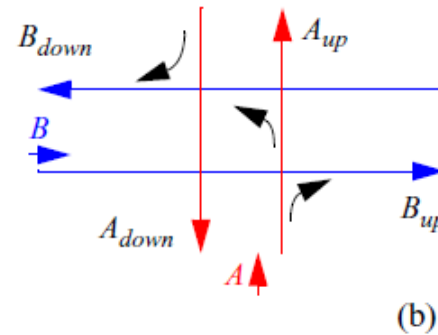
Moving Objects, Trajectories

- Road-Network Constrained

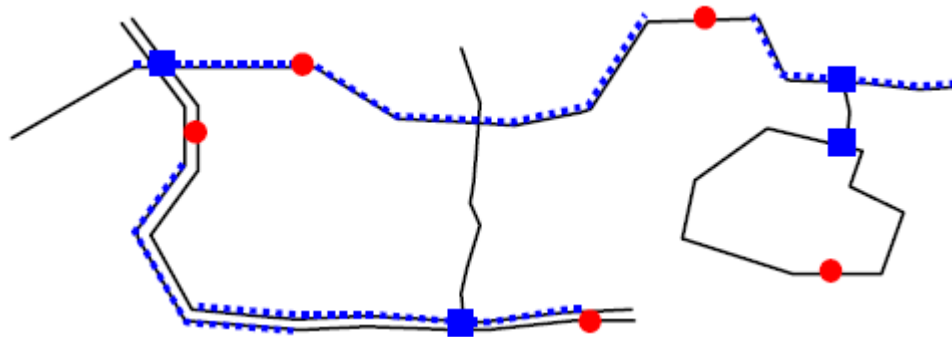


	A_u	A_d	B_u	B_d
A_u	1	0	1	1
A_d	0	1	0	1
B_u	0	0	1	0
B_d	0	0	0	1

(c)



[GAD06,CKS+09]



Distance $\rightarrow$ graph-distance
 $\Rightarrow$ Dijkstra-like algorithmic approaches

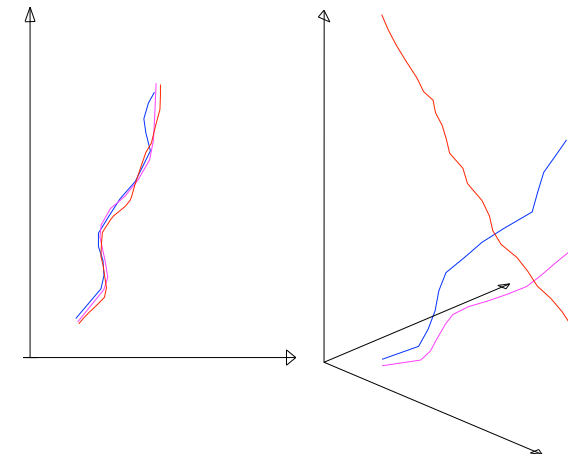
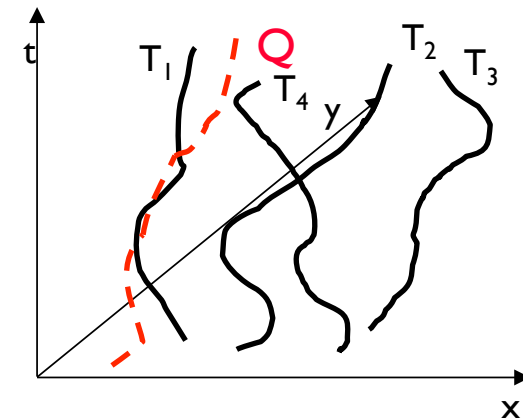
However, the flow/speed in a given link may vary in time...

Trajectory Similarity

- Given a query trajectory Q and a trajectory database $D=\{T_i\}$ search among them for the most similar trajectory
 - Determine the most similar by using not only the spatial shape of the trajectories but also their evolution in time

- Enhance the performance of the search by employing existing index structures (R-trees) used to support more traditional queries (range, NN e.t.c.).

[FGP+07] Frentzos, E., Gratsias, K., Pelekis, N., Theodoridis, Y.: Algorithms for Nearest Neighbor Search on Moving Object Trajectories. *Geoinformatica* 11(2): 159-193 (2007).



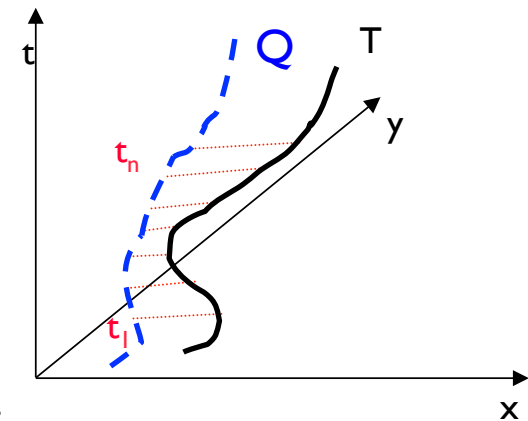
Metrics for Measuring Similarity

- Definition of Dissimilarity $DISSIM(Q,T)$ between trajectories Q and T being valid during the period $[t_1, t_n]$:
the definite integral of the function of time of the Euclidean distance between the two trajectories during the same period

$$DISSIM(Q,T) = \int_{t_1}^{t_n} D_{Q,T}(t) dt$$

- In real world, each trajectory is represented as a collection of discrete points

1. Apply linear interpolation in between
2. Use a discrete measure

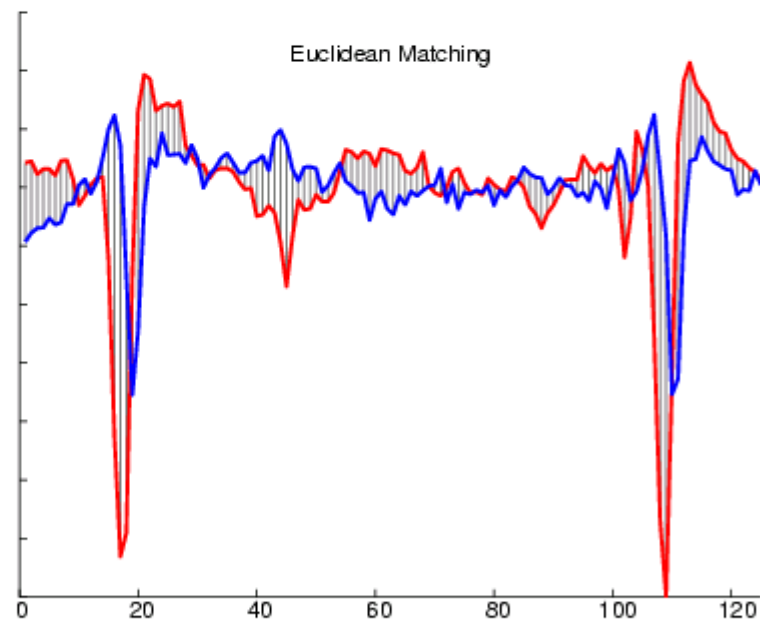


$$DISSIM(Q,T) = \sum_{k=1}^{n-1} \int_{t_k}^{t_{k+1}} D_{Q,T}(t) dt$$

Similarity Measures

- Euclidean (sensitive to time axis displacement)

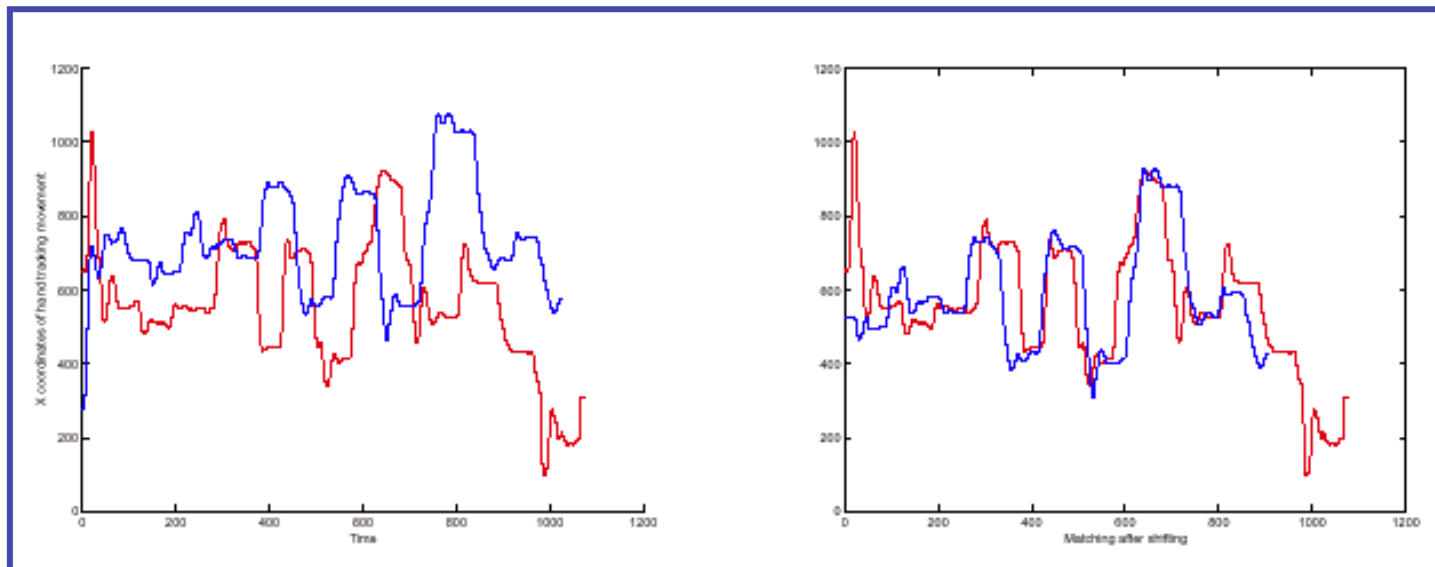
$$L_p\text{-Norm: } L_p = (\sum (x(i) - y(i))^p)^{1/p}$$



Requirements for Similarity Model

We need to address the following issues:

- Similar Motions in different space Regions



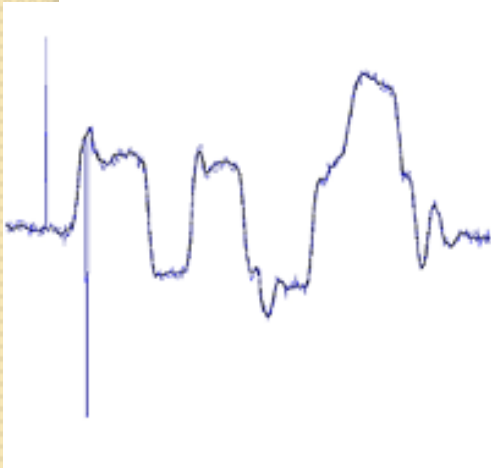
[VGK02] Vlachos, M., Gunopulos, D., Kollios, G.: Discovering Similar Multidimensional Trajectories. ICDE 2002

Requirements for Similarity Models

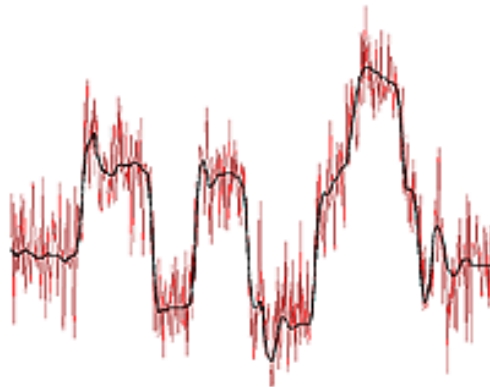
We need to address the following issues:

- Outliers

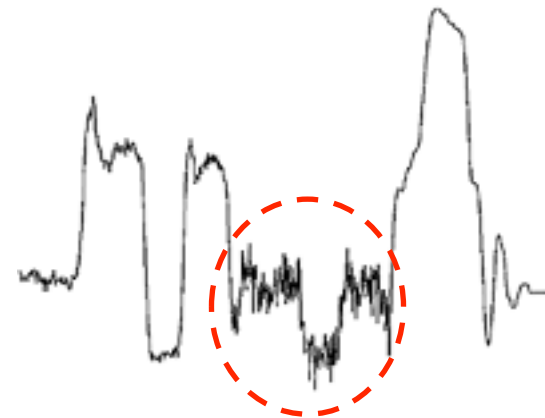
Random Peaks



Noise Everywhere



Non Recoverable Part



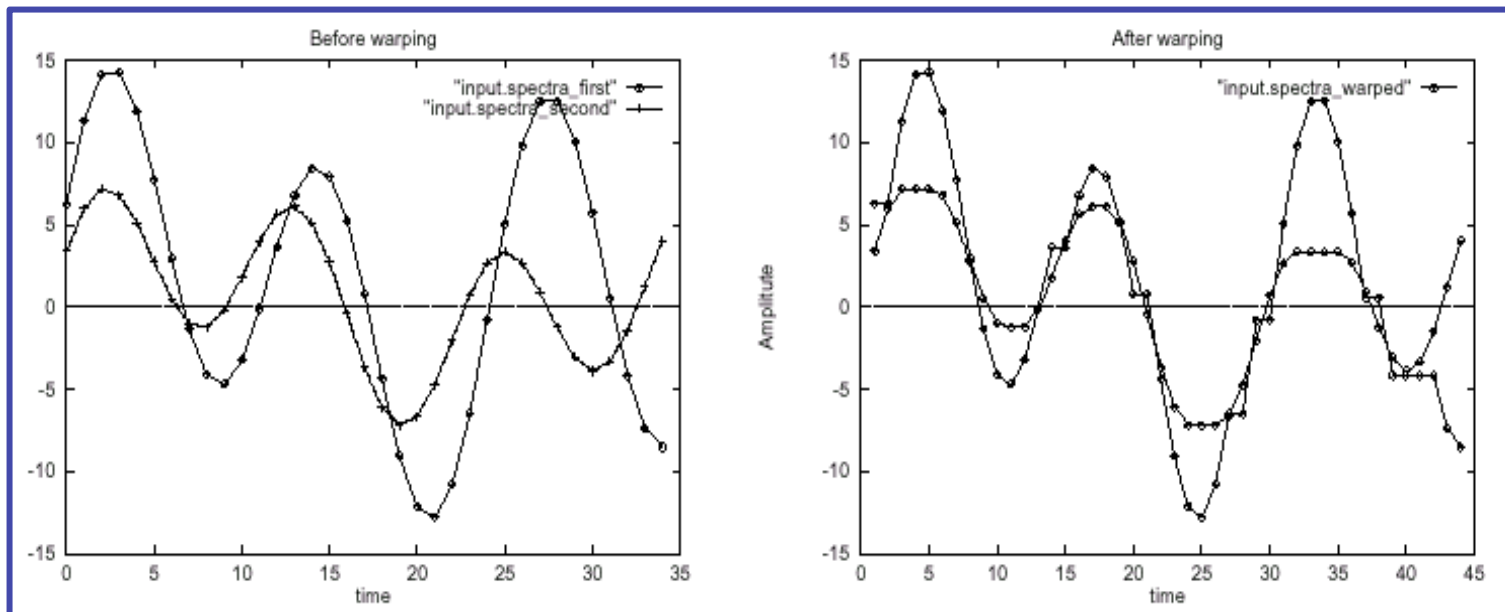
- Different Lengths

[VGK02]

Dynamic Time Warping (DTW)

- Time Warping
 - Allows stretching in time axis
 - Matches all elements
 - Extensively used in speech recognition domain

[VGK02]



Dynamic time warping (DTW)

- Euclidean (L2) distance is

$$D_2(\mathbf{x}, \mathbf{y}) := \sum_{t=1}^N (x_t - y_t)^2$$

or, recursively,

$$D_2(\mathbf{x}_{1:t}, \mathbf{y}_{1:t}) = (x_t - y_t)^2 + D_2(\mathbf{x}_{1:(t-1)}, \mathbf{y}_{1:(t-1)}).$$

- Dynamic time warping distance is

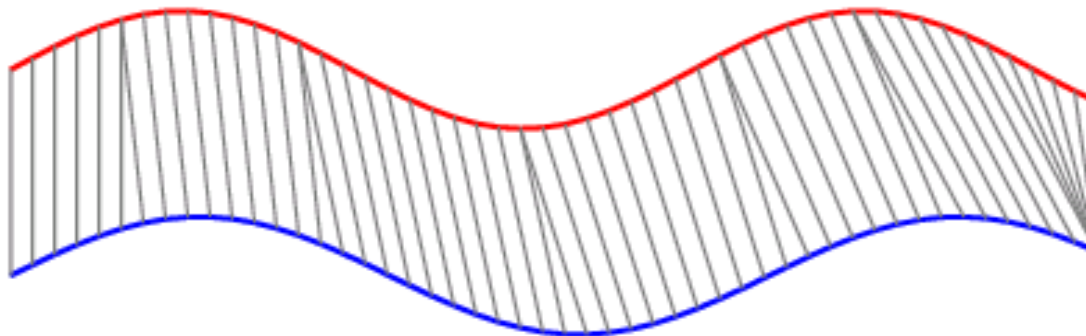
$$D_w(\mathbf{x}_{1:i}, \mathbf{y}_{1:j}) := (x_i - x_j)^2 +$$

$$\min \begin{cases} D_w(\mathbf{x}_{1:(i-1)}, \mathbf{y}_{1:(j-1)}) \\ D_w(\mathbf{x}_{1:(i-1)}, \mathbf{y}_{1:j}) \\ D_w(\mathbf{x}_{1:i}, \mathbf{y}_{1:(j-1)}) \end{cases}$$

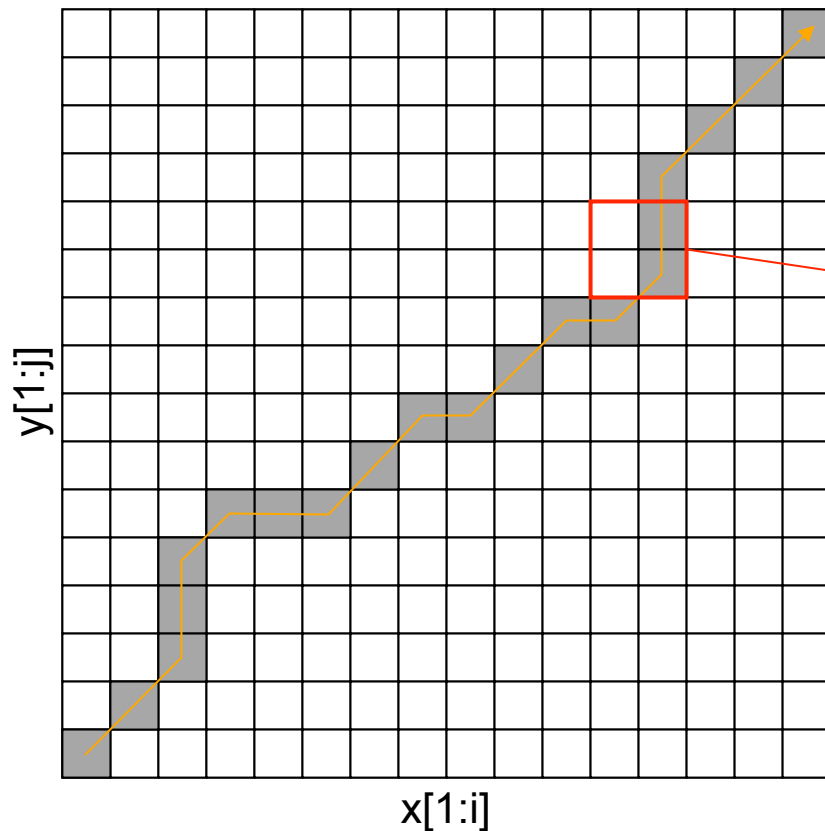
← shrink $\mathbf{x}$ / stretch $\mathbf{y}$

← stretch $\mathbf{x}$ / shrink $\mathbf{y}$

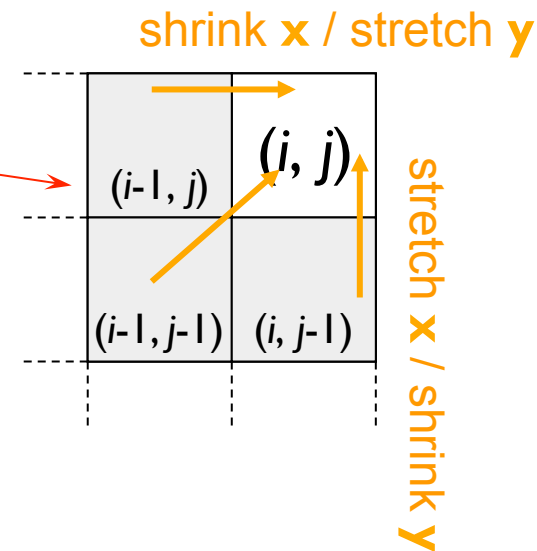
where $\mathbf{x}_{1:i}$ is the subsequence $(x_1, \dots, x_i)$



Dynamic time warping algorithm



- DTW allows any path
- Examine all paths:



- Standard dynamic programming to fill in table—top-right cell contains final result

[VP06] Vlachos, M., Papadimitriou, S.: Hands-on Time-Series Analysis with Matlab. Tutorial, ICDM 2006

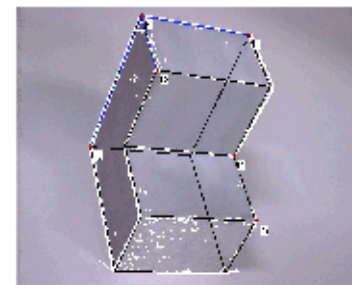
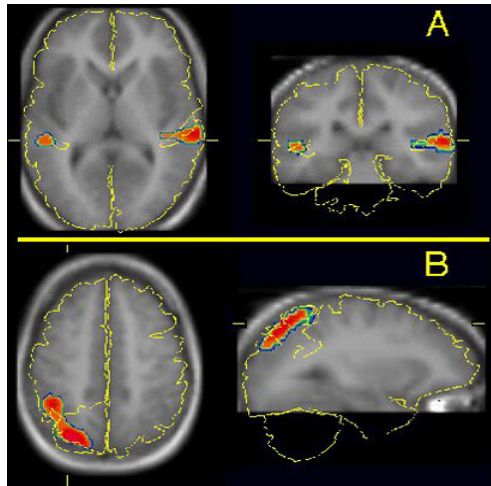
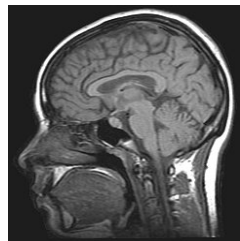
DTW – Acceleration

a) Constraints, b) Trajectory approximation, c) Indexing
Global Constraints

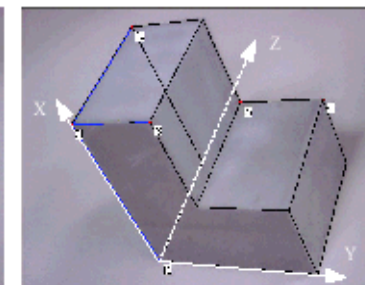
Find the optimal warping path within the band (region with gray):

- SCB (where $p_t = (i,j)_t$ and $j-R \leq i \leq j+R$)
- Itakura Parallelogram

Rotation Invariant Similarity

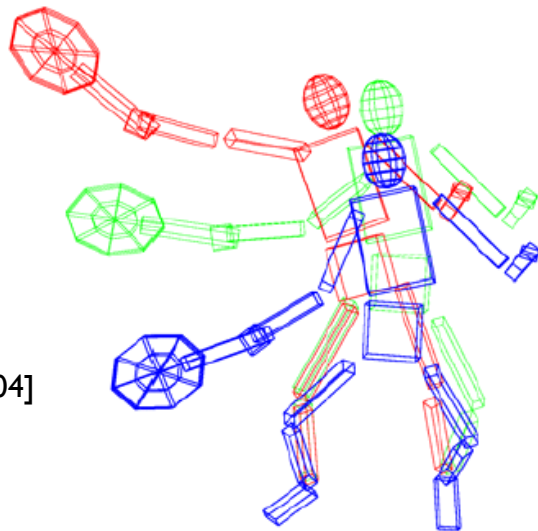


(a) Perspective view 1

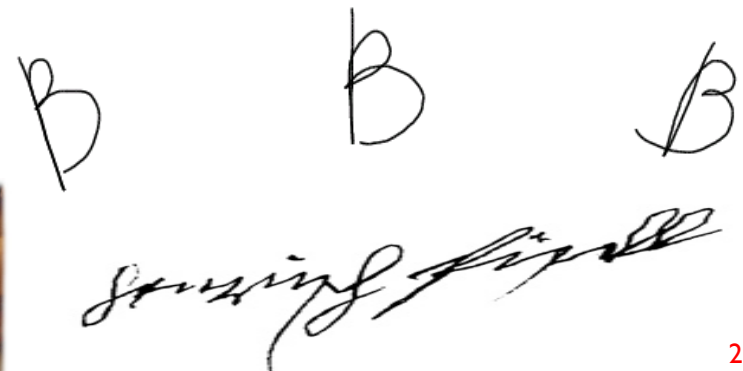


(b) Perspective view 2

- Robot vision (object contour recognition)
- Medical applications (CAT scan registration)
- Handwriting recognition (tablet PCs, signatures)
- Pose estimate –MOCAP data (video games, movies)



[VGD04]

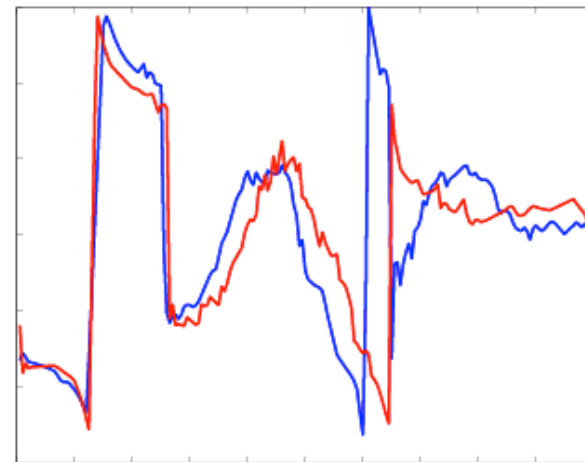
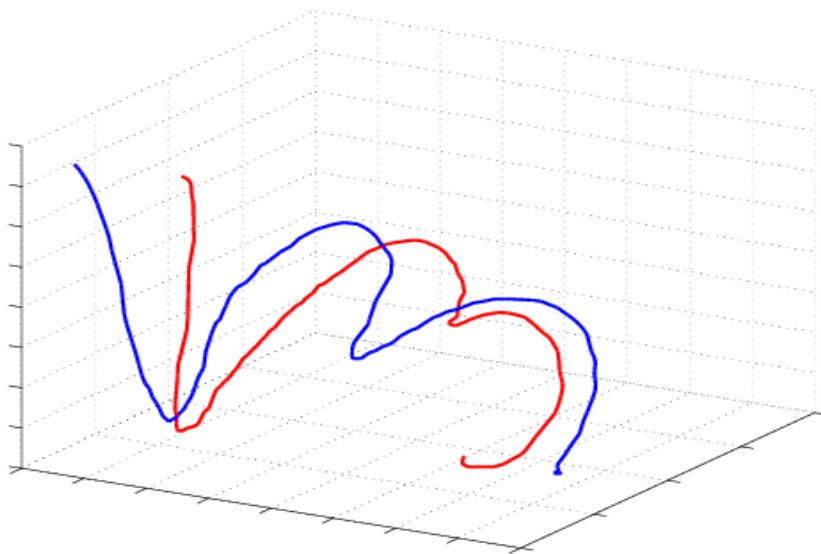


Space Transformation

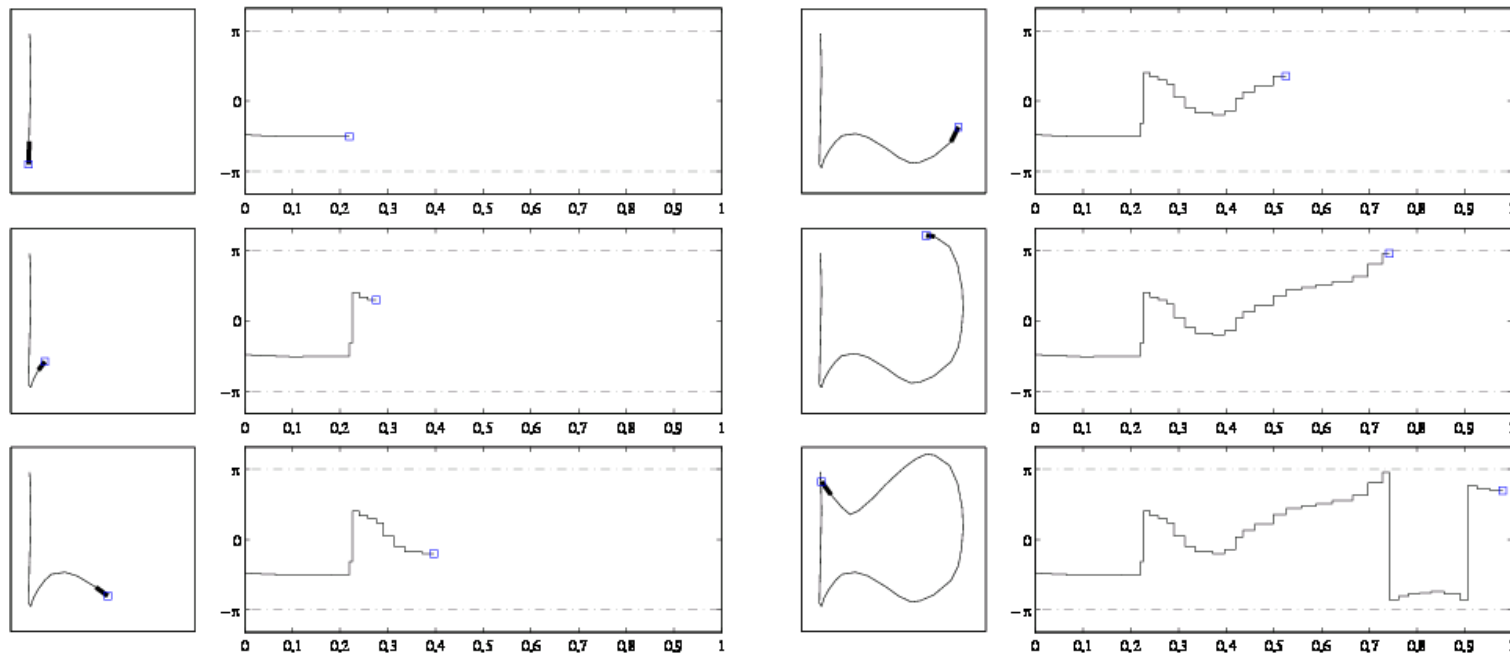
Method:

- Transform trajectory to rotation invariant space
 - *Positional information -> Angle/ Arc length information*
- Proper normalization ensures also:
 - *Translation invariance*
 - *Scale Invariance*
- Perform matching in new space
 - *Use Warped Matching to compensate for shape variations*

[VGD04] Vlachos, M., Gunopulos, D., Das, G.: Rotation invariant distance measures for trajectories. KDD 2004



Mapping Example



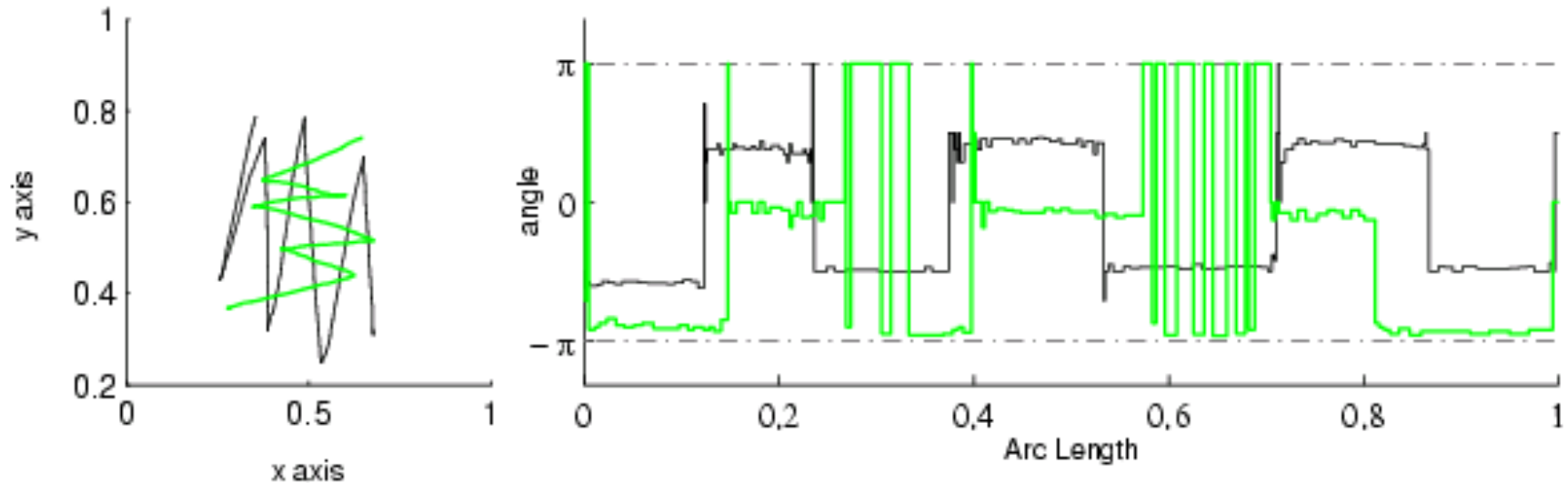
Angles to utilize:

- Exact angles (*as in figure*)
- Relative Angles (to previous segment)
- Angles from center of mass

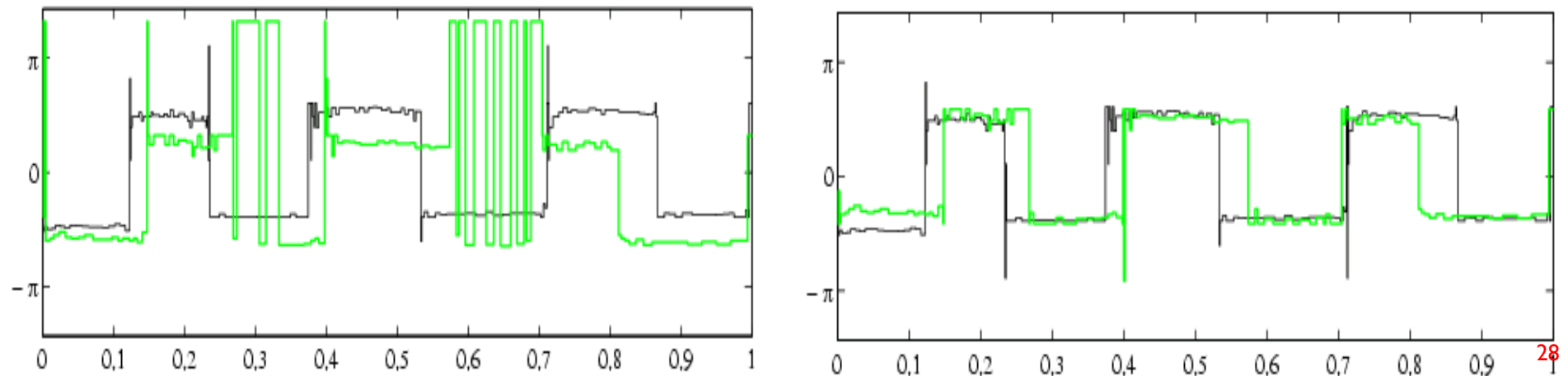
[VGD04]

Mapping Refinement (Vlachos et al 04)

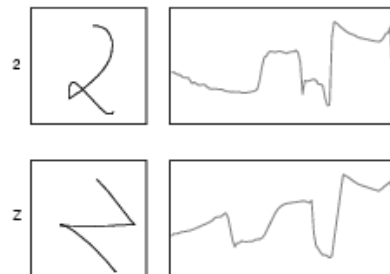
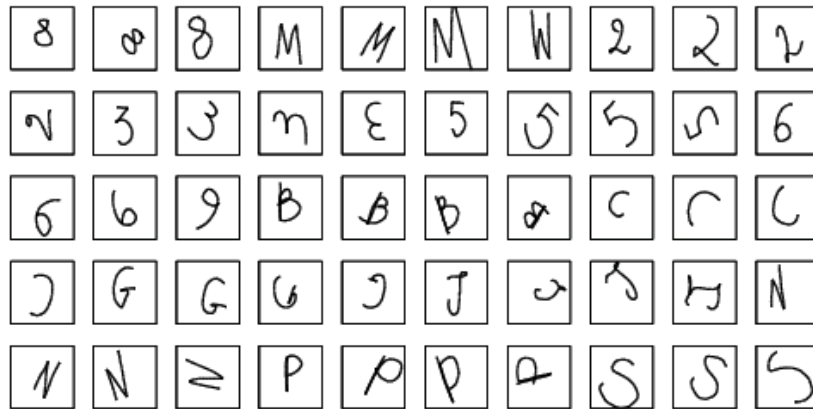
- Rotation by $\pi/2$ results in vertical shift by $\pi/2$



- Perform modulo- π iterative normalization in new space



Handwriting recognition example



[VGD04] **"Bad" Classifications**



2D Mapping according to new distance.

Rotated versions are mapped close in space

Longest Common Subsequence (LCSS)

Arithmetic Example:

- $t1=[0, 4, 6, 8, 7, 4, 6, 5, 6, 4, 6]$
- $t2=[0, 3, 4, 6, 7, 6, 3, 6, 4, 6]$

- **Definition**

$$LCSS(A, B) = \begin{cases} 1 + LCSS(\text{Head}(A), \text{Head}(B)), & \text{if } a_n = b_n \\ \max(LCSS(\text{Head}(A), B), LCSS(A, \text{Head}(B))), & \text{otherwise} \end{cases}$$

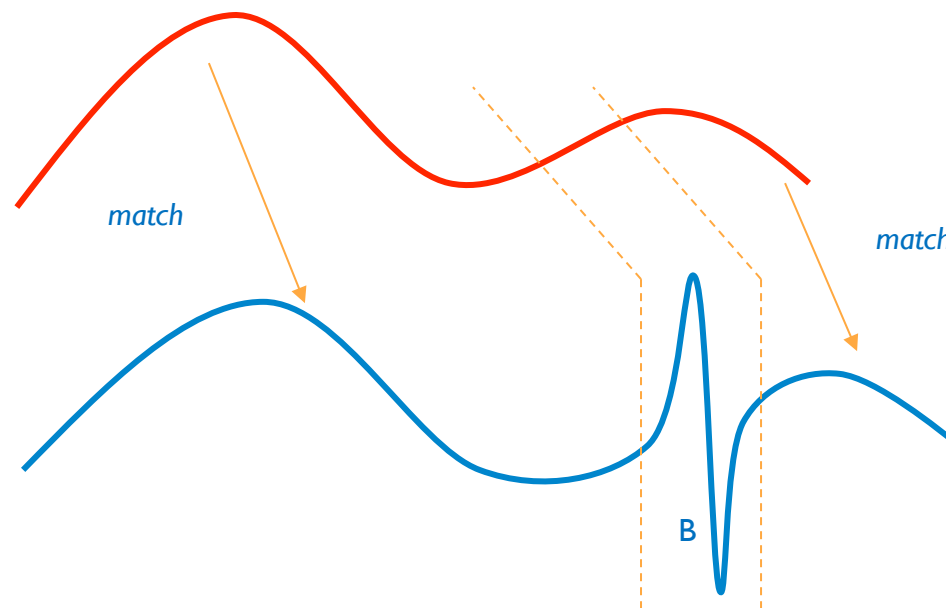
- **Dynamic Programming Solution**

$$LCSS[i, j] = \begin{cases} 0, & \text{if } i = 0 \\ 0, & \text{if } j = 0 \\ 1 + LCSS[i - 1, j - 1], & \text{if } a_i = b_j \\ \max(LCSS[i - 1, j], LCSS[i, j - 1]), & \text{otherwise} \end{cases}$$

Differences between DTW & LCSS

- DTW has to match every element, so is more sensitive to outliers
LCSS matches only the similar parts

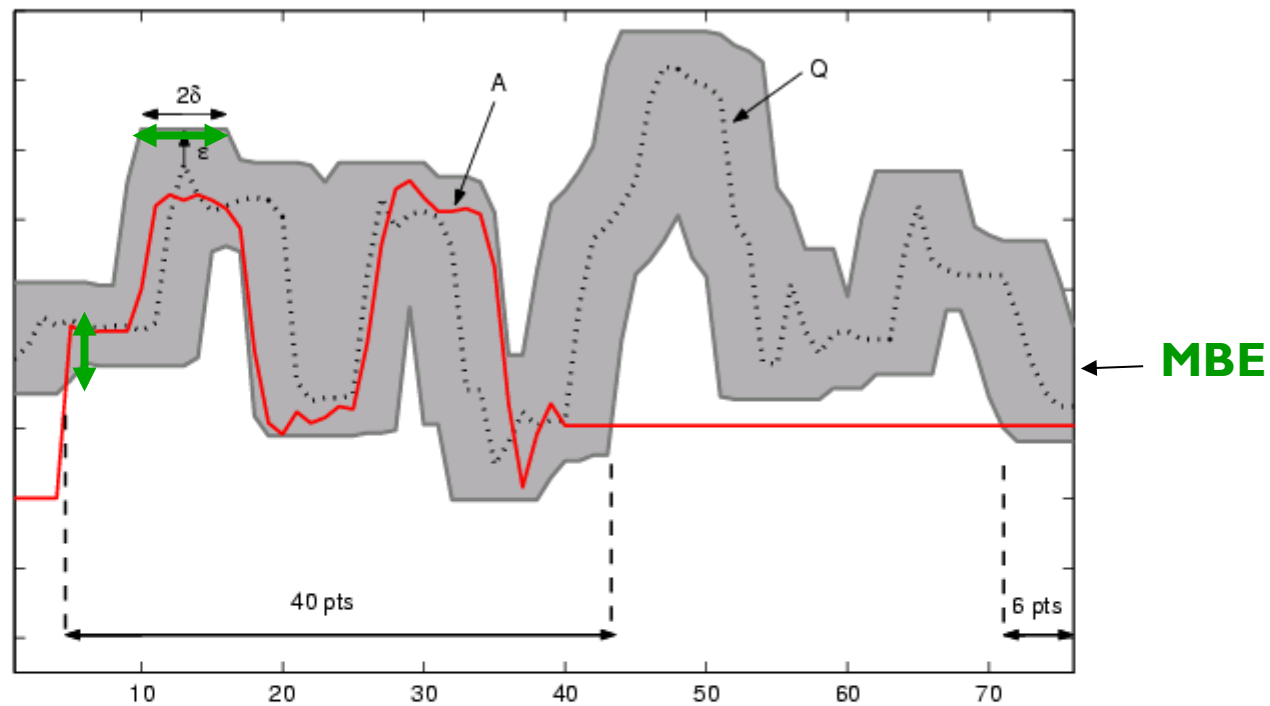
[VGK02]



[DTSWK08] provides experimental comparison of 8 representation methods; 9 similarity measures (+variants); 38 datasets. [WMDTSK12] provides further extensions.

LCSS for real values

$$LCSS_{\delta,\varepsilon}(A,B) = \begin{cases} 0 & \text{if } A \text{ or } B \text{ is empty} \\ 1 + LCSS_{\delta,\varepsilon}(\text{Head}(A), \text{Head}(B)) & \text{if } |a_n - b_n| < \varepsilon \text{ and } |n - m| \leq \delta \\ \max(LCSS_{\delta,\varepsilon}(\text{Head}(A), B), LCSS_{\delta,\varepsilon}(A, \text{Head}(B))) & \text{otherwise} \end{cases}$$

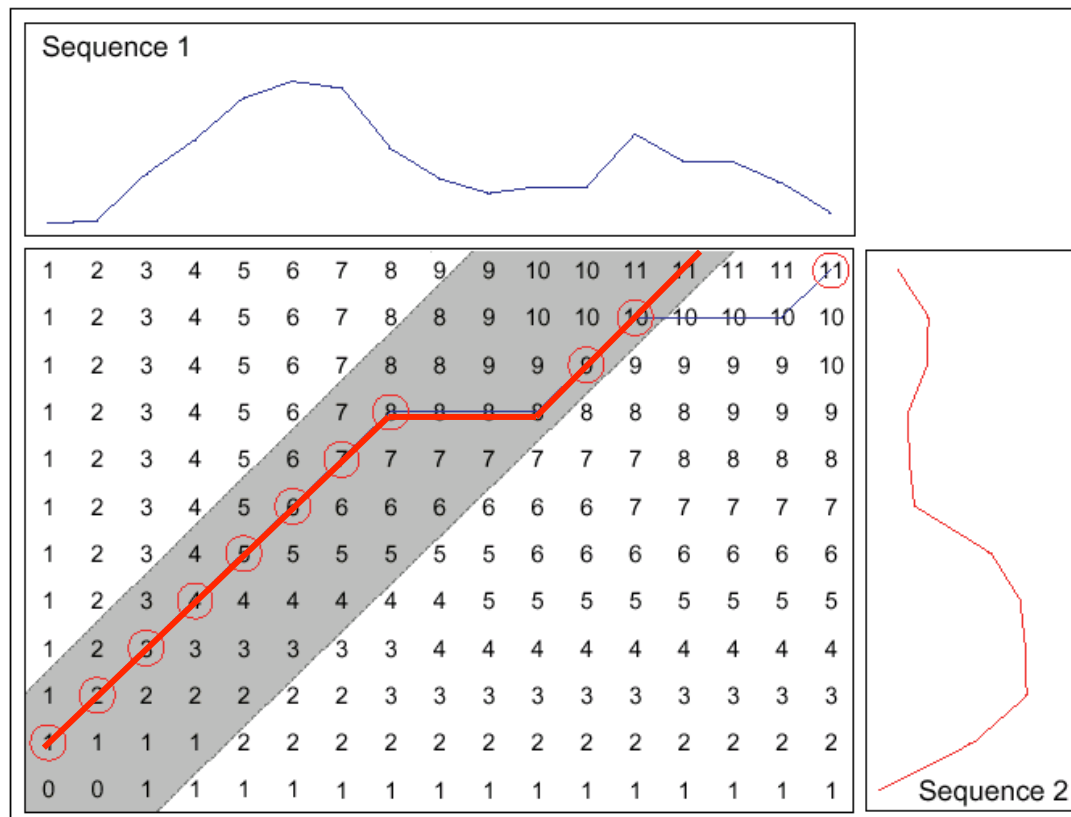


[VGK02]

LCSS Dynamic Programming algorithm

LCSS can be computed in $O(\delta(l_1 + l_2))$ by dynamic programming algorithm.

$$LCSS[i, j] = \begin{cases} 0 & \text{if } i = 0 \\ 0 & \text{if } j = 0 \\ 1 + LCSS[i - 1, j - 1] & \text{if } a_i = b_i \\ \max(LCSS[i - 1, j], LCSS[i, j - 1]) & \text{otherwise} \end{cases}$$



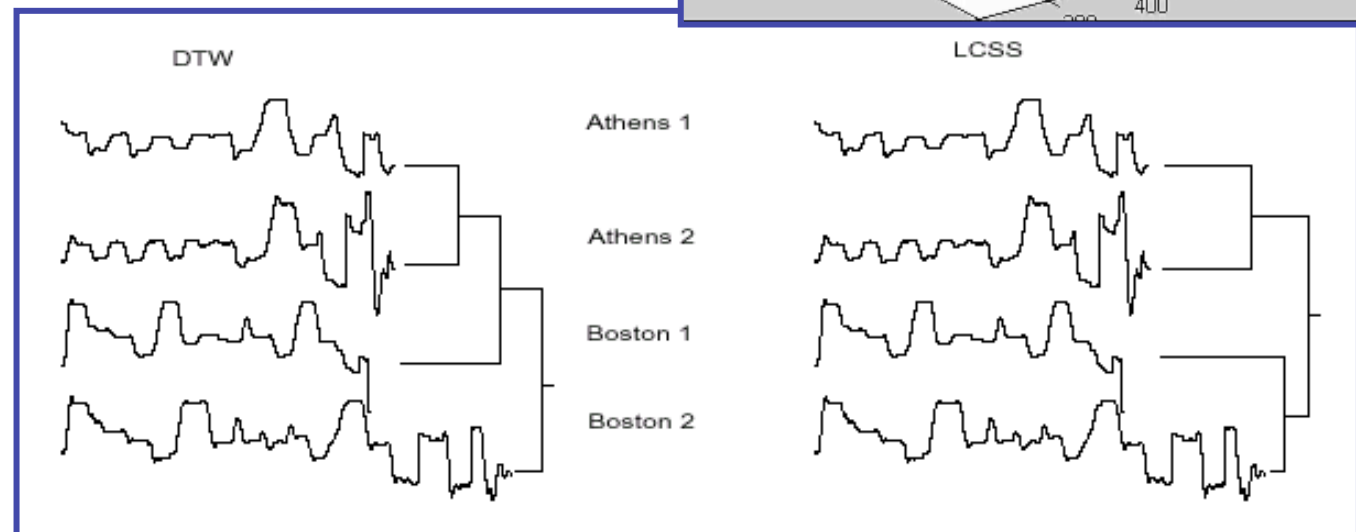
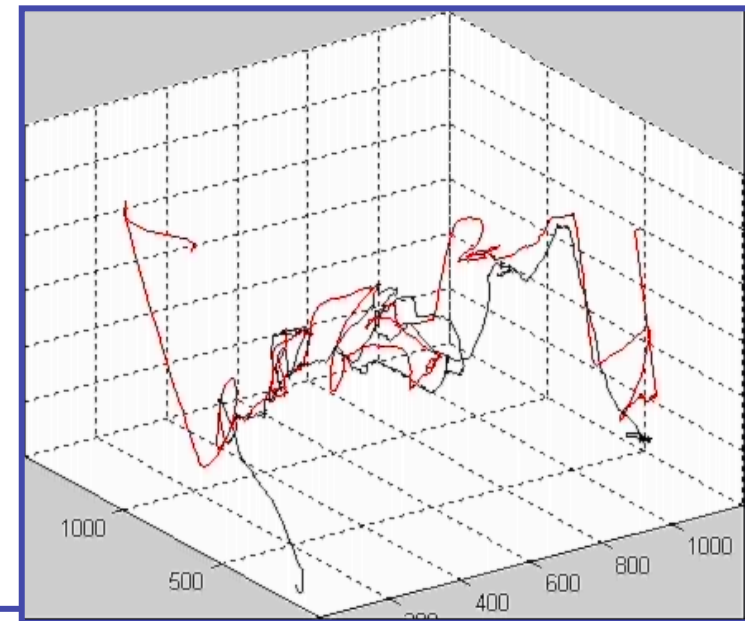
$$Sl(\delta, \epsilon, A, B) = \frac{LCSS_{\delta, \epsilon}(A, B)}{\min(n, m)}$$

Classification Accuracy

Example using Video Tracking data

- The 2D time series of a human tracking feature (e.g. the end of a finger). In conjunction with a “spelling program” the user can “write” various words.
- The data we used correspond to: ‘athens’, ‘berlin’, ‘london’, ‘boston’, ‘paris’.

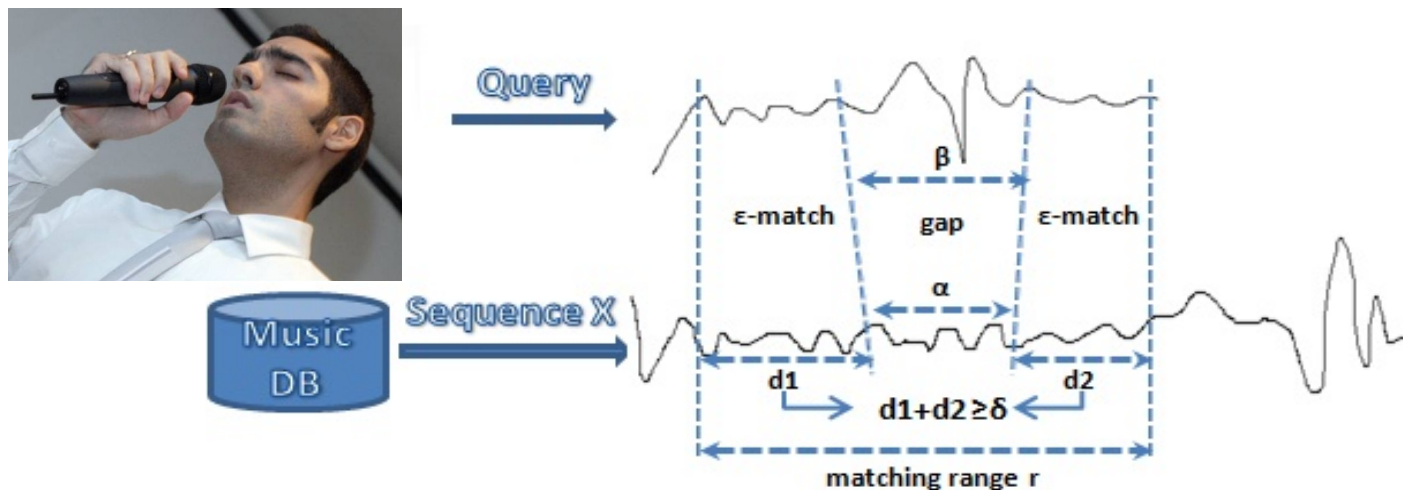
Test classification accuracy using Hierarchical Clustering



[VGK02]

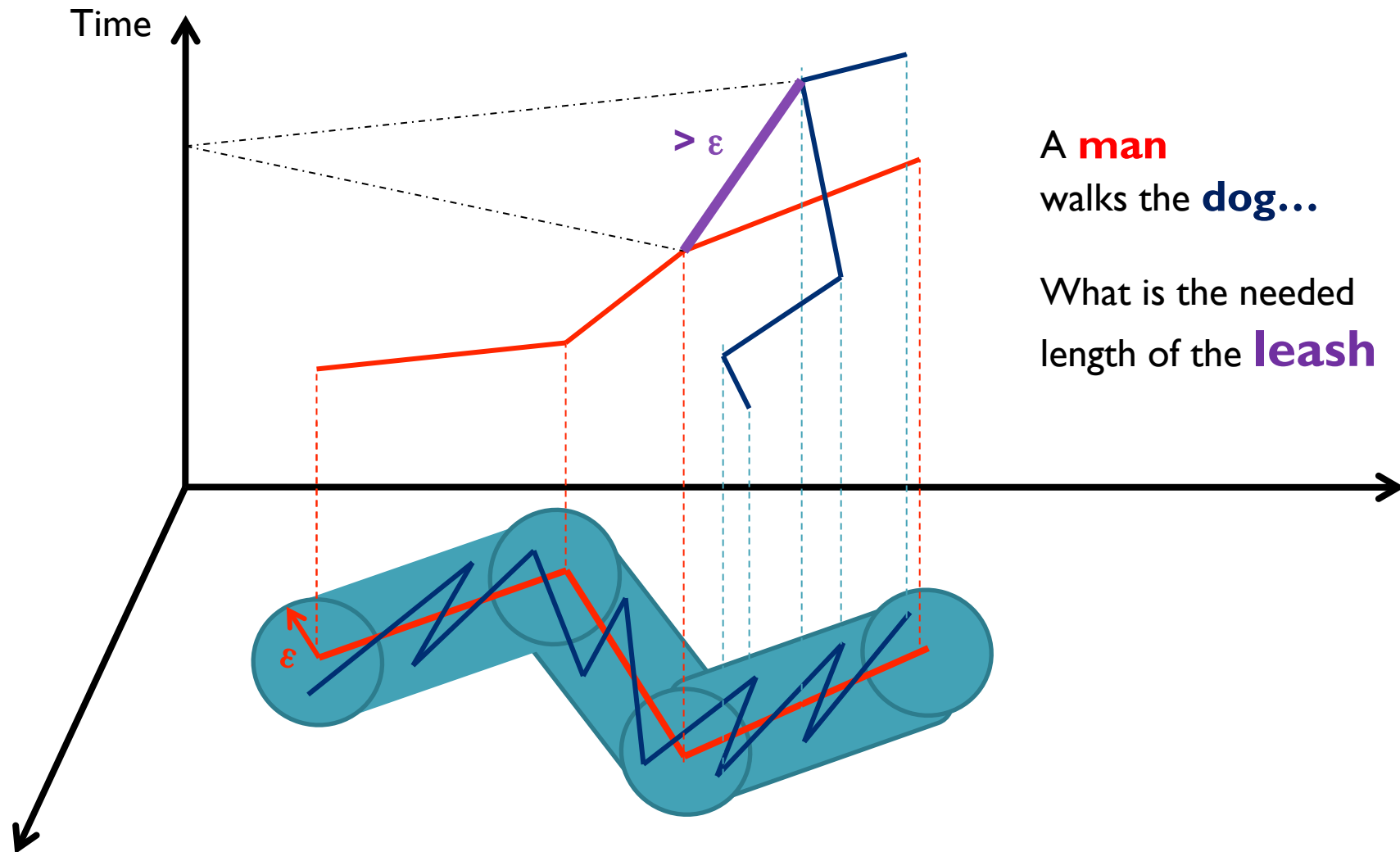
A variant of LCSS for querying by humming

Consider $Q[ts_1 : te_1]$ and $X[ts_2 : te_2]$ of equal length. Let G_Q, G_X the indices of those elements in Q and X that are included in the corresponding subsequences. If $q_{\pi_i} \approx_{\varepsilon} x_{\gamma_i}, \forall \pi_i \in G_Q, \forall \gamma_i \in G_X, i = 1, \dots, |G_Q|$, and $\pi_{i+1} - \pi_i < \beta, \gamma_{i+1} - \gamma_i < \alpha$, then pair $\{Q[ts_1 : te_1], X[ts_2 : te_2]\}$ defines a common gapped subsequence of Q and X . The longest such subsequence with $te_2 - ts_2 \leq r$ is called $SMBGT(Q, X)$.



SMBGT: error-tolerant matching denoted as ε -match.

- Distance Functions



A **man**
walks the **dog...**

What is the needed
length of the **leash**

- Fréchet distance (length of the leash)

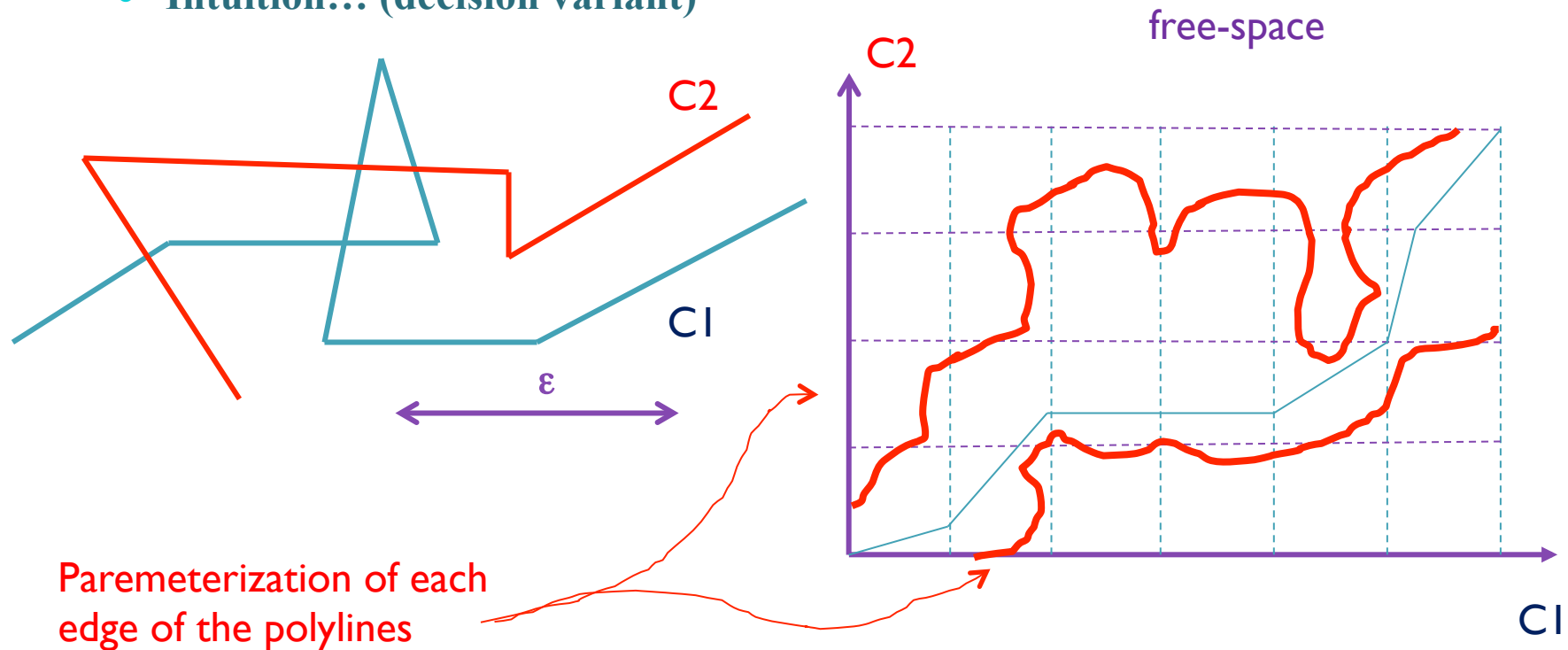
Let $C_1: [a_1, b_1] \rightarrow \mathbf{R}^2$ and $C_2: [a_2, b_2] \rightarrow \mathbf{R}^2$ denote two curves
their Fréchet distance is defined as

$$\delta_f(\mathbf{C}_1, \mathbf{C}_2) = \inf_{\alpha, \beta} \max_{t \in [0, 1]} \| \mathbf{C}_1(\alpha(t)) - \mathbf{C}_2(\beta(t)) \|$$

α (respectively β) range over all the continuous and monotonically increasing
functions $[0, 1] \rightarrow [a_1, b_1]$ (respectively $[0, 1] \rightarrow [a_2, b_2]$)

- **Fréchet distance** (length of the leash)

- Intuition... (decision variant)



Parameterization of each
edge of the polylines

$\delta_f(C_1, C_2) \leq \epsilon$ if there is a monotonically increasing path

Decision: $O(mn)$

Computation: $O(mn \log(mn))$ [AG95]

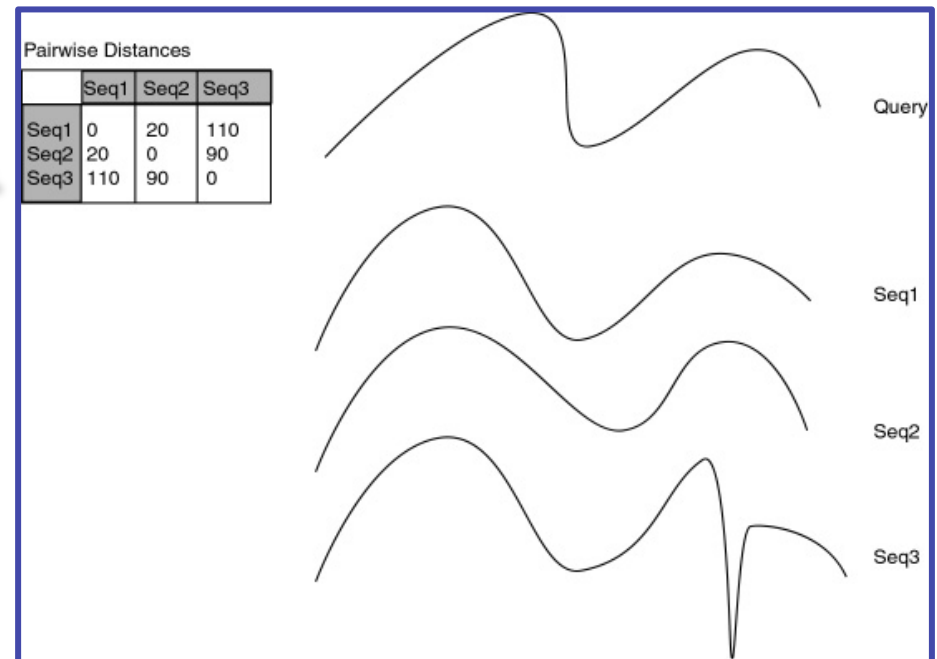
Note: this is for two fixed curves

Indexing using the Triangle Inequality

- Objective: Create an index to answer Nearest Neighbor Queries for new trajectories

Pruning power of triangle inequality:

- best-match-so-far = 20, $D(Q, S2) = 150$.
- $D(S2, S1) = 20$ and
 $D(Q, S1) \geq D(Q, S2) - D(S2, S1) \Rightarrow$
 $D(Q, S1) \geq 150 - 20 = 130$ we can prune S1.
- Similarly, we can prune S3.



Metric Properties of Similarity Measures

Triangle inequality does not hold for DTW or LCSS
(but holds for Edit Distance or DTW with gaps).

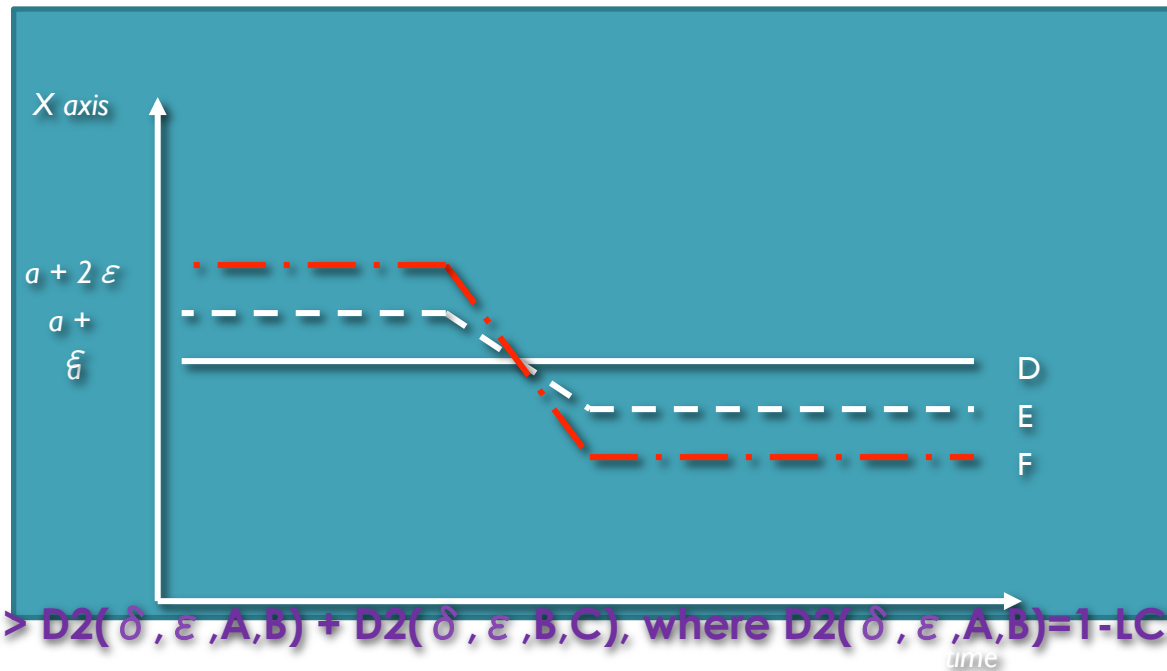
Example 1: $A = 1,1,5$

$B = 2,2,5$

$C = 2,2,2,2,2,5$

Then: $DTW(A,B) = 2$, $DTW(B,C) = 0$, $DTW(A,C) = 5$

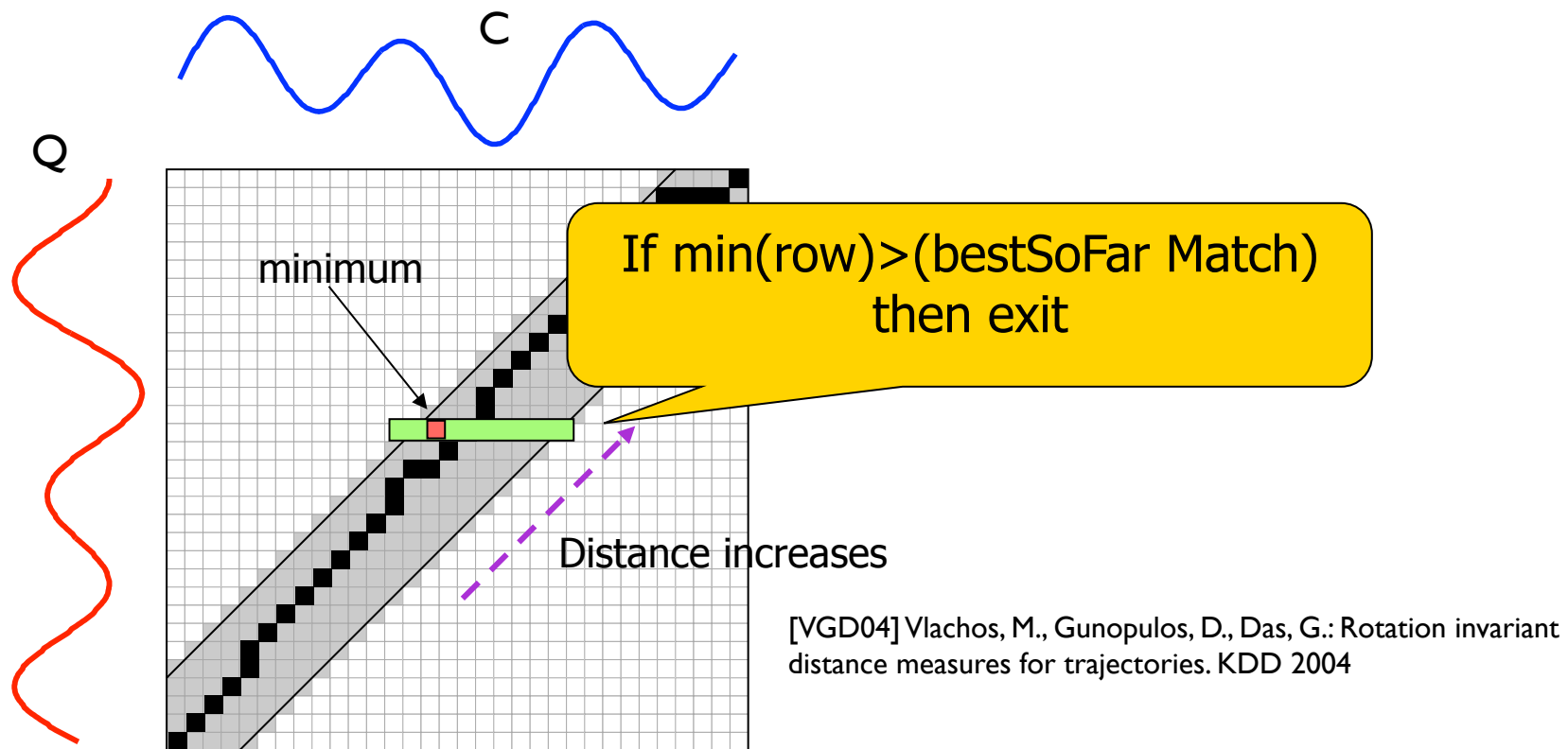
Example 2:



$D1(\delta, \epsilon, A, C) > D2(\delta, \epsilon, A, B) + D2(\delta, \epsilon, B, C)$, where $D2(\delta, \epsilon, A, B) = 1 - LCSS(\delta, \epsilon, A, B)$.

Sequential Scan

- Compare the Query to all the trajectories in the database
- Use Lower/Upper Bounds or Early Termination to speed up the search:

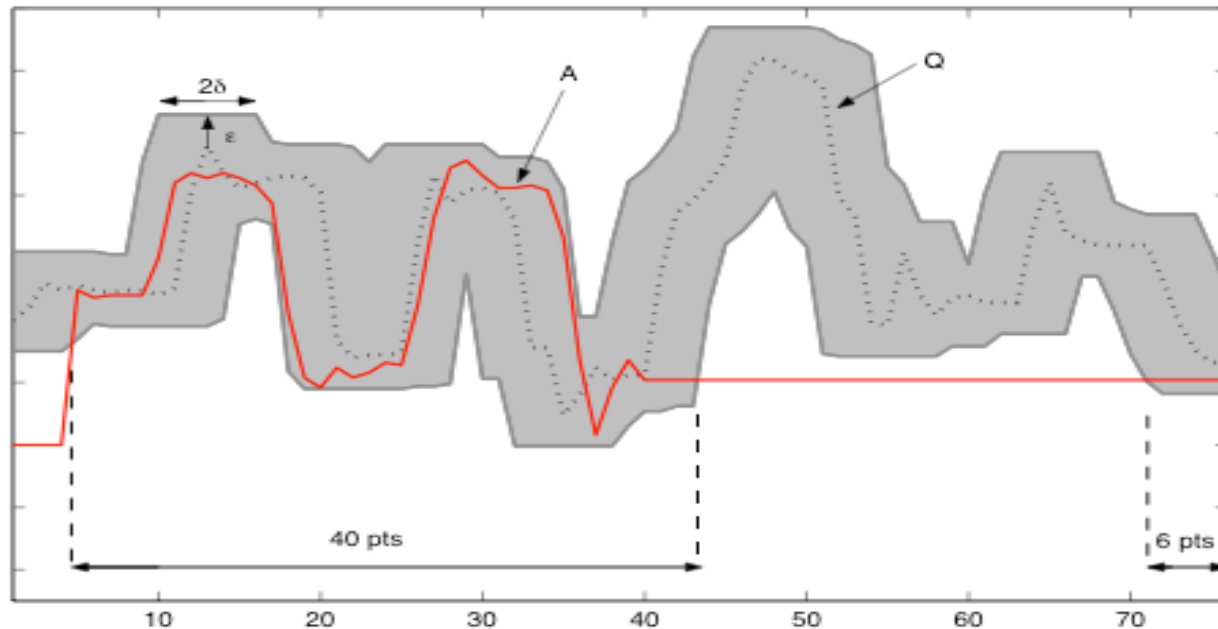


Computing Bounds: The Envelope Approach

$$EnvLow \leq MBE_{\delta, \epsilon}(Q) \leq EnvHigh, \text{ where } \begin{cases} EnvHigh[i] = \max \{Q[j] + \epsilon\}, & |i - j| < \delta \\ EnvLow[i] = \min \{Q[j] - \epsilon\}, & |i - j| < \delta \end{cases}$$

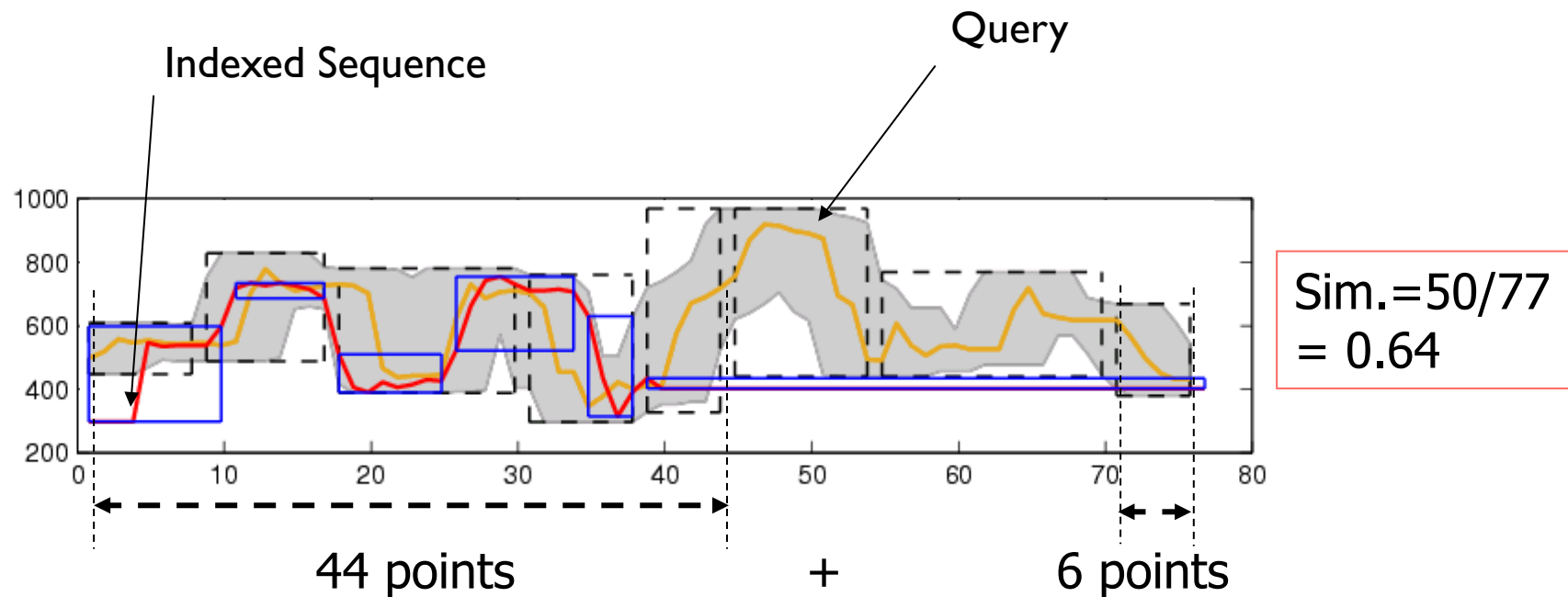
$$LCSS_{\delta, \epsilon}(MBE(Q), A) = \sum \begin{cases} 1, & \text{if } A[i] \text{ within envelop} \\ 0, & \text{otherwise} \end{cases}$$

Theorem: $LCSS_{\delta, \epsilon}(Q, A) \leq LCSS_{\delta, \epsilon}(MBE(Q), A)$



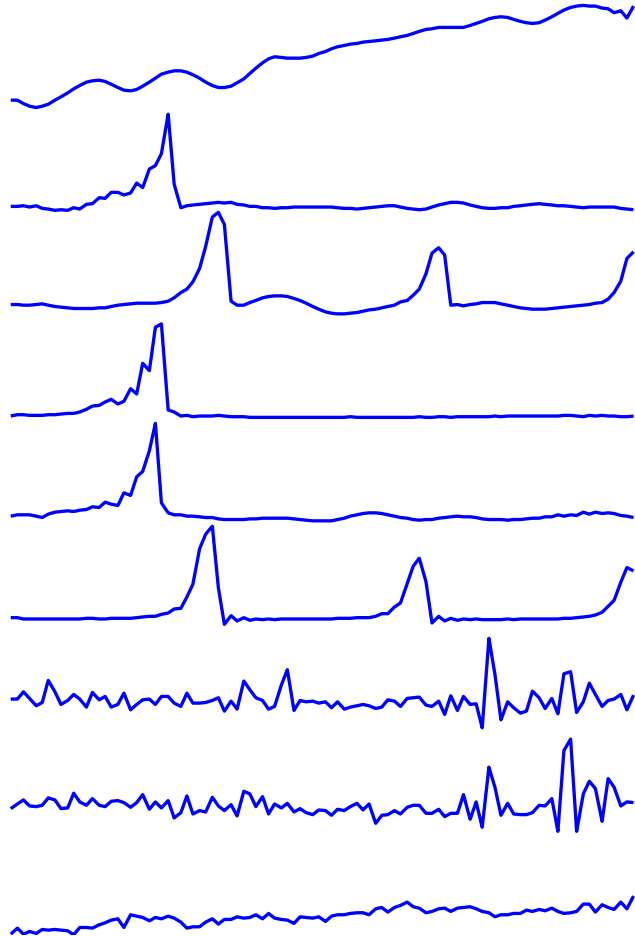
LCSS Upper Bound (Vlachos et al 03)

- LCSS captures similarity
- $\text{Similarity} = 1 - \text{dist}$

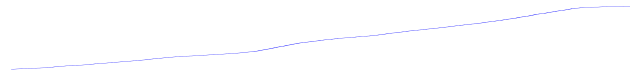


Using bounds for filtering: Example

sequences

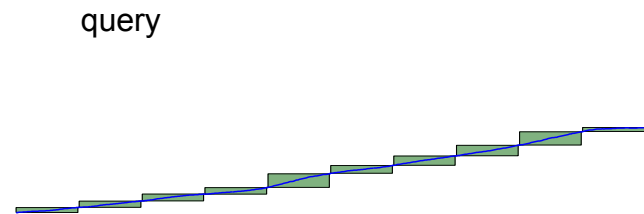
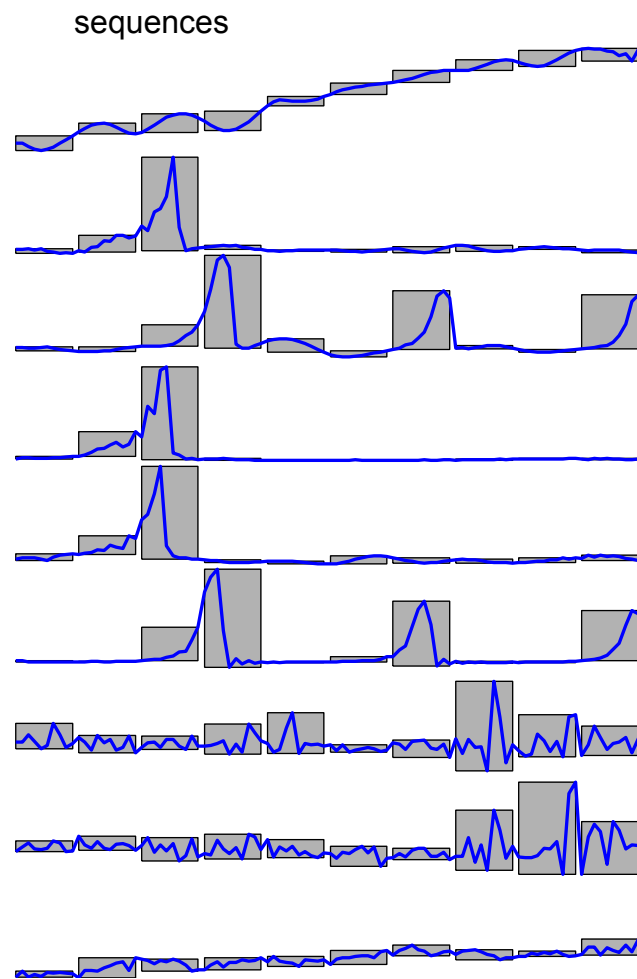


query



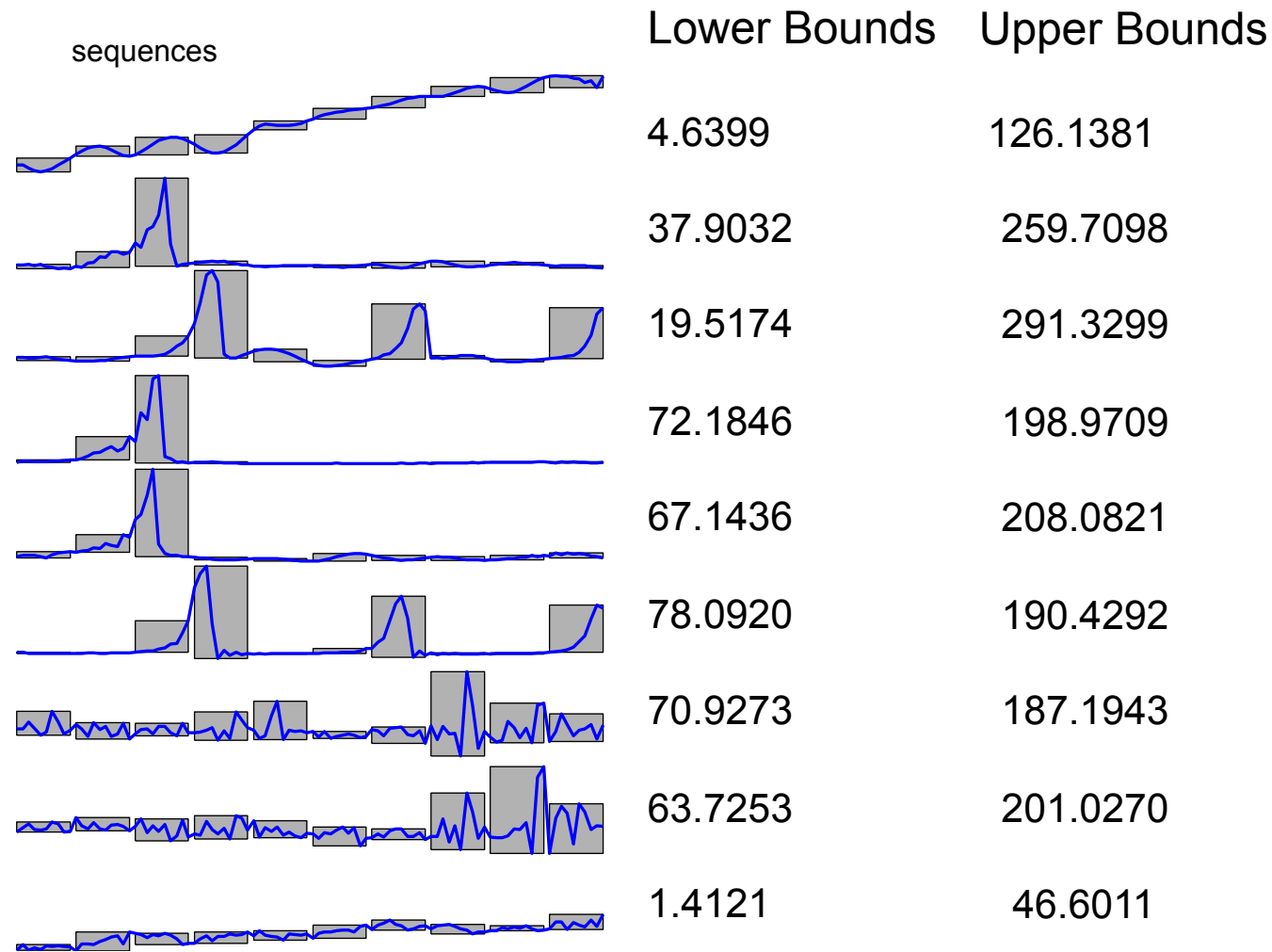
[VHG+03]

Using bounds for filtering: Example



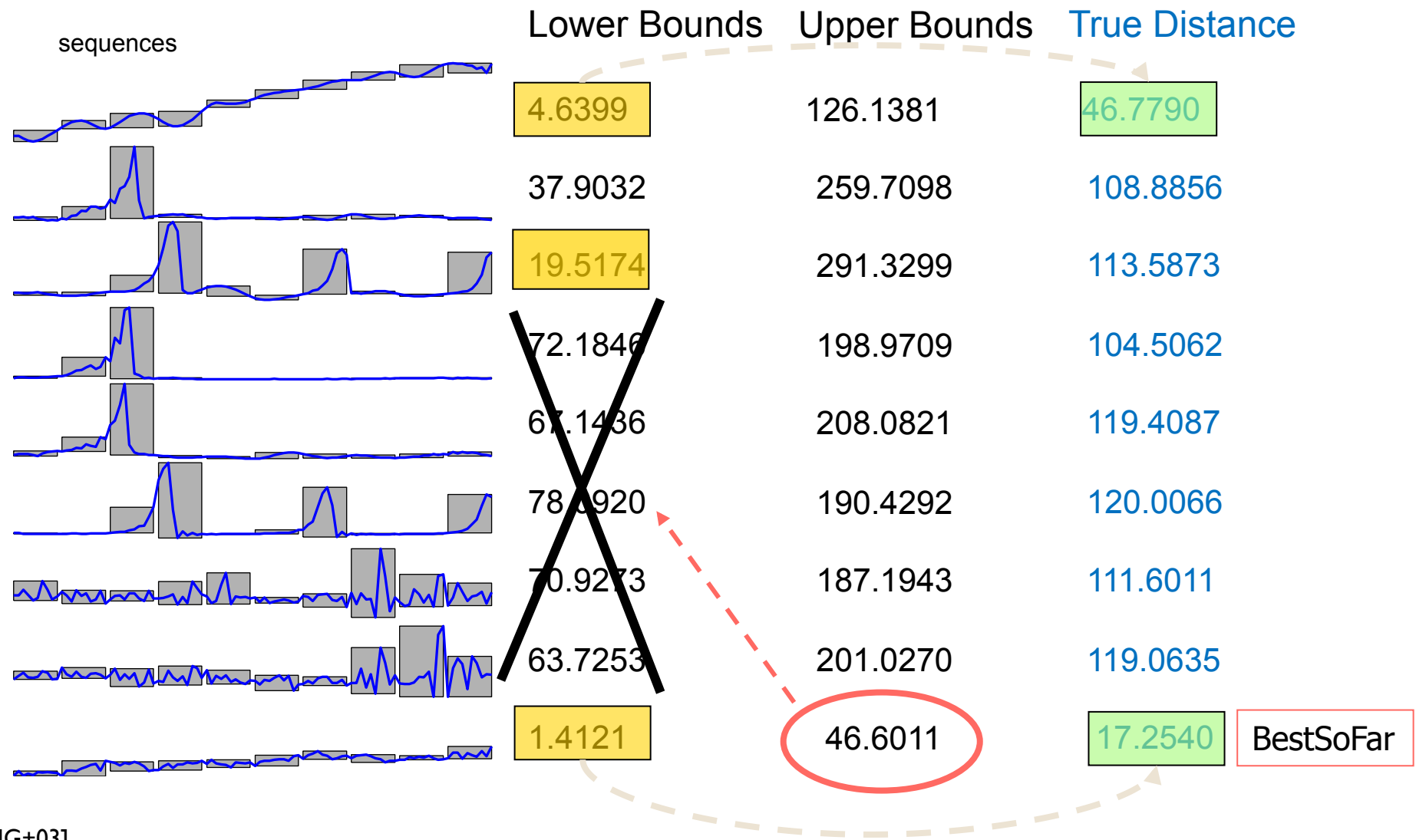
[VHG+03]

Using bounds for filtering: Example



[VHG+03]

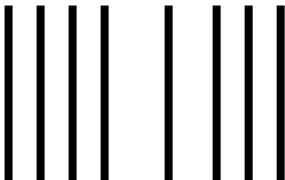
Using bounds for filtering: Example



Subsequence matching

- Example: Biological sequences (DNA)
- Near homology search:
 - Deviation from Q does not exceed a threshold $\leq 15\%$

Q: TCTAGGGCA



...ACTTAGCTGTAGTCGT**TCTATGGCA**TATGCATGCTGATCTCGTGCGTATG...

The Edit Distance [Levenshtein et al. 1966]

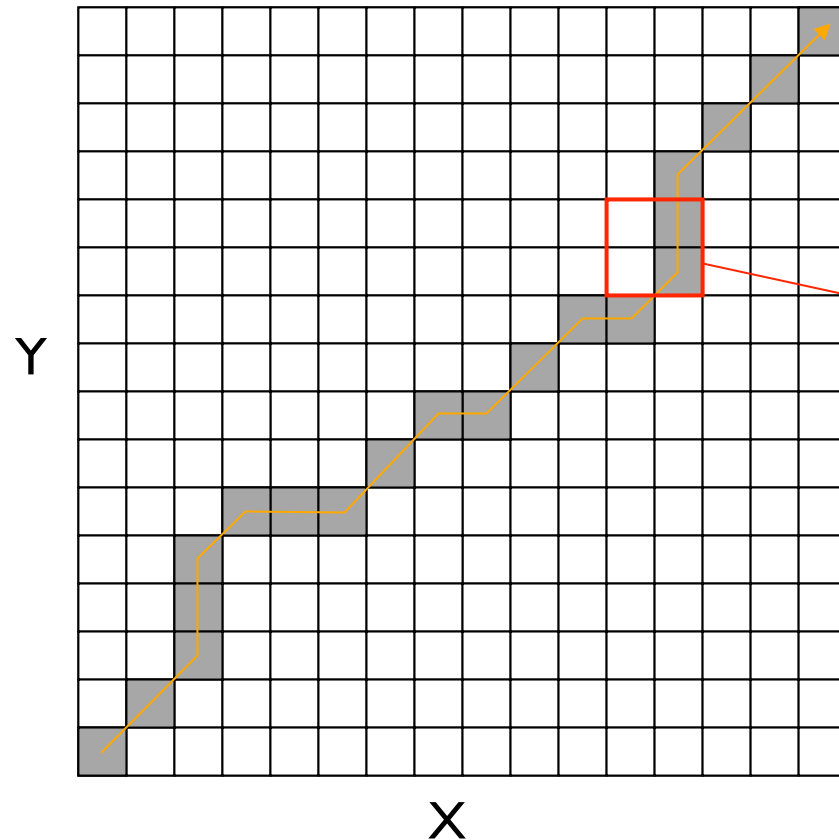
- Measures how dissimilar two strings are
- $ED(A,B)$ = minimum number of operations needed to transform A into B
- Operations = [insertion, deletion, substitution]
- Well known dynamic programming quadratic algorithm
- Example:

• $A = ATC$ and $B = ACTG$

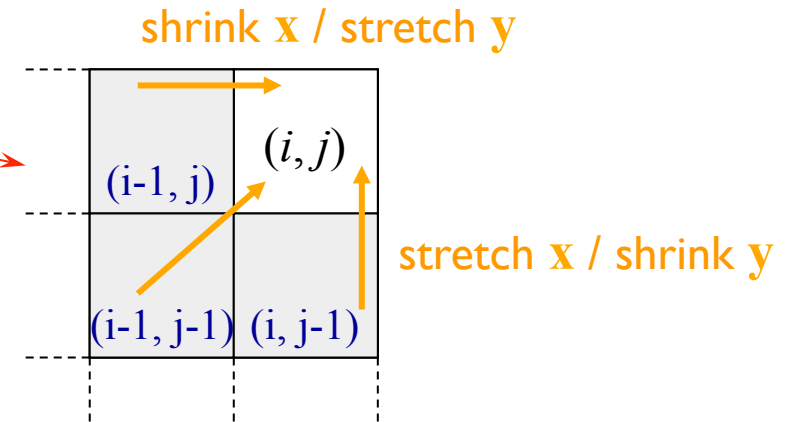
$A = A - T C$
 | . | .
 $B = A C T G$

$ED(A,B) = 2$

Edit Distance Dynamic Programming Algorithm



- matching allows any path.
- Examine all paths:

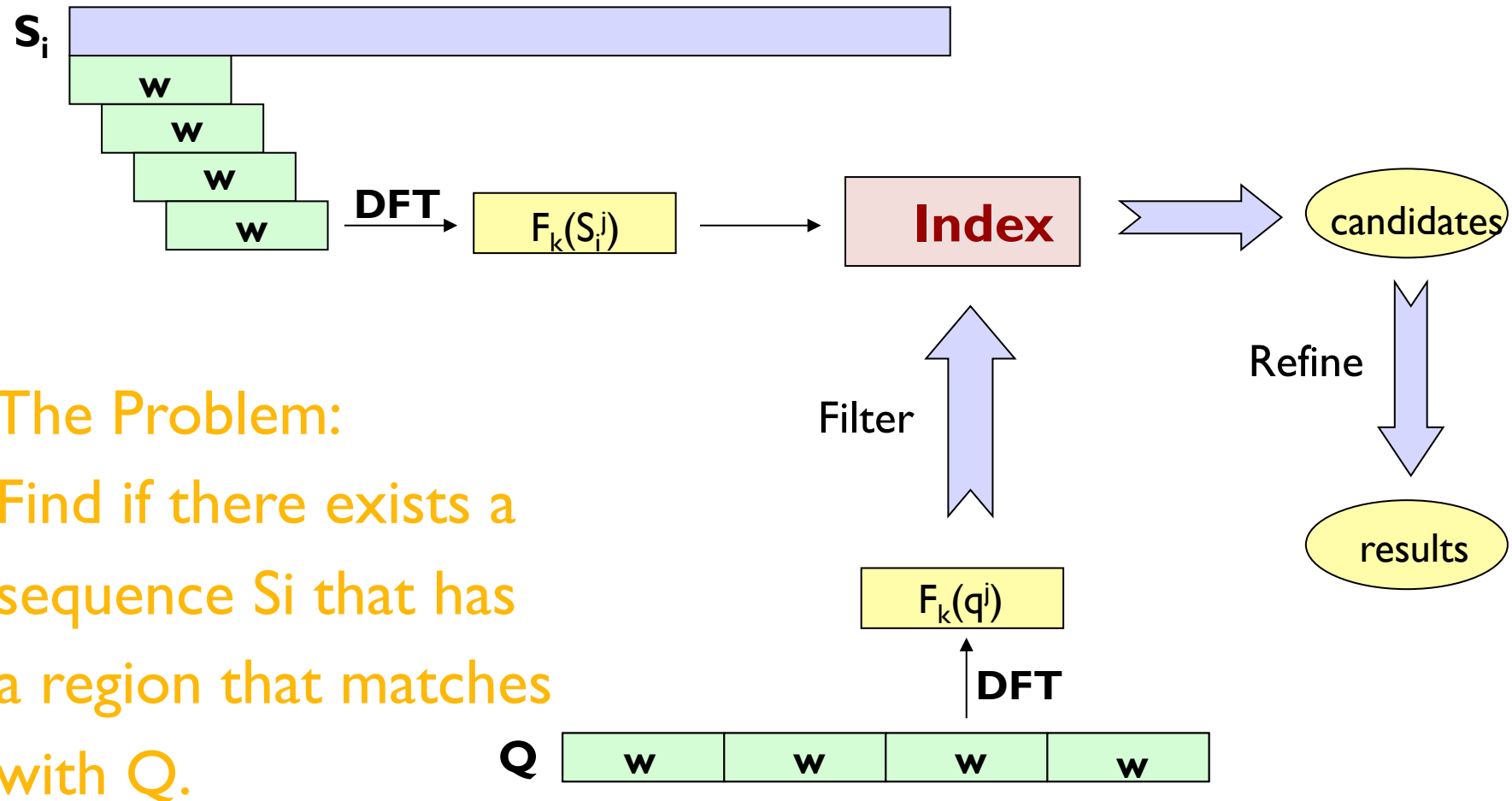


- Standard dynamic programming to fill in the table.
- The top-right cell contains final result.

Smith-Waterman [Smith&Waterman et al. ('81)]

- Similarity measure used for local alignment:
 - Match can be a *subsequence* of the query sequence.
- Define penalties:
 - match, mismatch, gaps (insertion and deletion).
 - Scoring parameters are defined by the user.
- Example:
 - $A = \text{ATCA}$ and $B = \text{TATTCG}$
 - match = 2, mismatch = -1, gaps = -1.

ST-Index: Subsequence Matching Using the Euclidean Distance (Faloutsos et al, 94)



[FRM94] Christos Faloutsos, M. Ranganathan, Yannis Manolopoulos: Fast Subsequence Matching in Time-Series Databases. SIGMOD Conference 1994

Thanks and Acknowledgements

- Goce Trajcevski (EDBT'12 tutorial co-author)
- Michalis Vlachos and Spiros Paparimitriou (check their excellent tutorial on time series analysis using Matlab)
- Mohamed Mokbel (tutorials on Location-Aware Query Processing and Privacy in Mobile Data); Somayeh Dodge; Aris Goulalas.
- Gennady Adrienko, Vassilis Athitsos, Gautam Das, Hui Ding, Fosca Giannotti, George Kollios, Alex Kotsifakos, Bart Kuijpers, Song Lin, Bart Moelans, Apostolos Papadopoulos, Nikos Pelekis, Yucel Saygin, Peter Scheuermann, Dragan Stojanovic, Roberto Tamassia, Yannis Theodoridis, Dennis Vaccaro, George Valkanas, Marcos Vieira, Xiaoyue Wang, Demetrios Zeinalipour-Yazti

**Research supported by the NSF-CNS and EU
INSIGHT EU FP7 project: www.insight-ict.eu
DISFER, MMD, INCEPTION, GeomComp projects**



NORTHWESTERN
UNIVERSITY



European Union
European Social Fund



MINISTRY OF EDUCATION & RELIGIOUS AFFAIRS, CULTURE & SPORTS
MANAGING AUTHORITY

Co-financed by Greece and the European Union



ΕΘΝΙΚΟΝ & ΚΑΠΟΔΙΣΤΡΙΑΚΟΝ
ΠΑΝΕΠΙΣΤΗΜΙΟΝ ΑΘΗΝΩΝ
NATIONAL & KAPODISTRIAN
UNIVERSITY OF ATHENS